

The First Break: Career Effects of Initial Access Shocks in Tournament Labor Markets*

Hugo Sant’Anna[†]

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Abstract

Does a worker’s *first* randomly assigned opportunity shape careers more than subsequent ones? I study lucky losers in professional tennis, qualifying-round losers who enter the main draw when another player withdraws, and restrict the analysis to first-time recipients so that both treated and control players are LL-naive. Grand Slam allocation is randomized among the top four qualifying losers ($N = 120$ ATP, $N = 82$ WTA); a non-Grand-Slam control-function analysis extends the design to larger samples ($N = 2,054$ ATP; $N = 1,423$ WTA). Ranking-point gains peak at 12 weeks on the ATP tour (+82.8; Fisher $p = 0.028$) and at 8 weeks on the WTA tour (+55.0; $p = 0.017$), imply about \$100,000 in prize money over the subsequent 26 weeks, and fade by 52 weeks as the rolling ranking window absorbs new performance signals. On the ATP tour the mechanism is access rather than ability (Elo is unchanged at every horizon); the WTA tour shows modest Elo gains that align with an additional ability channel, and marginal effects diminish with LL experience (Wald tests reject equality of first- and second-LL effects on non-GS access; ATP $p = 0.002$; WTA $p = 0.009$). A verified-lottery subsample built from a data-only criterion supports the ATP 26-week effect ($p = 0.050$) and the WTA 8-week effect ($p = 0.005$), so ranking-based contamination is not driving the headline result.

JEL Codes: J24, J44, L83

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*I thank seminar participants at Emory University for helpful comments. Replication code and data are available at <https://github.com/hugosantanna/tennis-11>. All errors are my own.

[†]Department of Economics, Emory University. Email: hugo.santanna@emory.edu.

1 Introduction

Does a worker’s *first* random access shock matter more than subsequent ones? In tournament-style labor markets, ranking determines access and access determines the opportunity to earn further ranking, so an initial break can trigger a cascade that later opportunities cannot replicate. This paper studies lucky losers in professional tennis: players who lose in the final round of qualifying but enter the main draw when another player withdraws. At Grand Slam tournaments, lucky losers are selected by lottery among the top four ranked qualifying losers, giving a natural experiment in which otherwise similar players are randomly granted or denied a main-draw opportunity. I restrict the analysis to players who have *never previously received* a lucky loser slot, isolating the effect of a first-ever access shock on career trajectories. The setting supports clean measurement of both tournament access and playing ability, a separation that is difficult to obtain in other labor markets.

The mechanism behind career propagation in this setting is path dependence through the ranking system. Professional tennis uses a rolling 52-week ranking that determines direct entry into tournament main draws. A player who earns ranking points at one event may gain direct entry to subsequent events, generating further opportunities to earn points. Access generates exposure, exposure generates accumulation, and accumulation generates future access, forming a feedback loop. A one-time exogenous shock to ranking points can trigger this cascade if the initial shock is large enough. Players who receive a lucky loser slot but lose in the first round earn minimal ranking points and trigger no cascade. Players who win one or more main draw matches earn enough points to cross direct-entry thresholds at lower-tier events, setting off a sequence of additional opportunities. For a player who has never before received this type of opportunity, the first break is potentially pivotal: subsequent opportunities arrive after the player has already crossed direct-entry thresholds, so their marginal value is smaller.

The empirical strategy has three components. First, Grand Slam lucky loser allocation is random within the top-four pool, providing the primary identification. I restrict the sample

to first-time LL candidates (players with no prior LL awards of any type), so both treated and control players are LL-naive. I estimate immediate effects at the focal event and dynamic outcomes at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks, tracing the trajectory of access and ability from the immediate aftermath through one year. Second, I extend the analysis to non-Grand-Slam events using a control function approach: the generalized residual is computed from the exact enumeration probability that a player’s rank among qualifying losers falls within the top slots, a probability determined by the full combinatorial structure of qualifying match pairings. The non-Grand-Slam sample is also restricted to first-time LL candidates. Third, a match-level tournament performance model estimates whether lucky loser entry changes subsequent win probability conditional on opponent strength, again exploiting the Grand Slam lottery for identification.

In the first-time ATP Grand Slam sample ($N = 120$; 61 treated, 59 controls), the stacked specification shows that lottery selection raises ranking points by 72.5 at 4 weeks ($p = 0.008$), 82.8 at 12 weeks ($p = 0.002$), and 72.6 at 26 weeks (clustered $p = 0.047$; Fisher $p = 0.164$), fading by 52 weeks. These point estimates are roughly twice the effects estimated on the full sample that pools first-time and repeat LL recipients, consistent with the first break being pivotal. On the WTA tour ($N = 82$), effects concentrate at 8 weeks: +55.0 ranking points ($p = 0.017$). The first-time WTA sample also shows Elo gains at 8 weeks (+15.0, $p = 0.082$) and 52 weeks (+36.8, $p = 0.055$), a pattern absent from the pooled-LL WTA results and consistent with genuine ability improvements on the women’s tour. Non-Grand-Slam estimates using the control function approach extend these findings on larger samples ($N = 2,054$ ATP; $N = 1,423$ WTA): ATP ranking-point gains of 20.6 at 4 weeks ($p = 0.041$) and 44.6 at 26 weeks ($p = 0.002$), together with ATP Elo improvements at 8–26 weeks and a WTA Elo improvement at 12 weeks (+10.5, $p = 0.011$). A robustness analysis comparing the marginal effect of the n th lucky loser opportunity shows that first-time effects dominate: players receiving their second or third LL slot show weaker and often insignificant gains.

This paper makes four contributions. First, I provide the first causal estimates of how a

first within-career opportunity shock affects trajectories in a tournament labor market. The initial conditions literature has established that graduating in a recession (Oreopoulos et al., 2012), initial job placement (Oyer, 2006), and starting environment (Chetty et al., 2014) shape long-run outcomes. I test how a marginal within-career shock, specifically the first such shock, propagates through a transparent tournament market where performance is continuously measured and rankings update weekly. In the settings studied in the initial conditions literature effects persist for a decade or more; here, the access gains fade within a year as the rolling ranking window replaces the initial shock. The roughly twofold increase in point estimates relative to the pooled-LL sample shows that averaging over first-time and repeat recipients attenuates the effect of the pivotal first break; the fadeout indicates that persistence depends on how quickly the market incorporates new performance signals. Second, the access-versus-ability decomposition shows a more nuanced picture than simple “access begets access.” While ATP results align with the access channel, WTA first-timers show Elo gains, and non-Grand-Slam first-timers on both tours show ability improvements at medium horizons; a first break can generate learning that subsequent breaks do not replicate. Third, the marginal impact analysis provides direct evidence on diminishing returns to opportunity: Wald tests reject equality of first-LL and second-LL effects for tournament access ($p = 0.002$ ATP; $p = 0.009$ WTA in the non-Grand-Slam sample), and third-plus LL recipients on the WTA tour have negative ranking-point effects at longer horizons. Fourth, the match-level model shows that despite ranking-point gains, first-time ATP GS LL recipients do not outperform controls on a per-match basis: $\hat{\delta} = -0.141$ (player-clustered $p = 0.17$), directionally consistent with overseeding when novice entrants face opponents above their ability level, though the small first-time GS sample cannot separate this from zero. Ranking-point gains therefore come from more draws entered and more matches played rather than a per-match performance improvement. Even so, the access cascade delivers substantial dollar value: the implied total prize-money effect of a first-time Grand Slam LL entry is about \$100,000 USD over the subsequent 26 weeks, meaningful against qualifying-level players’ annual earnings

of \$40,000–\$80,000.

Section 2 reviews the related literature. Section 3 describes the institutional setting and data. Section 4 presents the econometric framework and identification strategies. Section 5 reports the main results. Section 6 discusses mechanisms. Section 7 presents robustness checks, including the marginal-impact analysis. Section 8 concludes.

2 Related Literature

This paper connects to the literature on career dynamics and path dependence. Oyer (2006) shows that initial labor market conditions have persistent effects on the careers of economists, with initial placement shaping long-run research productivity and job quality. Oreopoulos et al. (2012) reports that graduating during a recession reduces earnings for up to a decade. Oyer (2008) shows that stock market conditions at the time of MBA graduation determine whether graduates enter investment banking, with lasting income consequences. Chetty et al. (2014) documents geographic variation in intergenerational mobility, so the environment in which a person starts shapes the long-run trajectory. The common thread is that initial conditions matter because they shape access to subsequent opportunities, a form of cumulative advantage (Merton, 1968). This paper tests whether the same logic extends to within-career opportunity shocks that are randomly assigned, and separates the access and ability dimensions of the response.

A second strand examines the returns to marginal opportunities and access shocks. The closest labor market analog to the lucky loser setting is Pallais (2014), who shows that random provision of a single work opportunity on oDesk leads to persistent gains in subsequent employment, mirroring the access-begets-access mechanism documented here. Abel et al. (2020) report that randomly assigned reference letters improve labor market outcomes for disadvantaged job seekers in South Africa. Startz (2021) shows that random interviewer assignment affects employment outcomes, so small procedural shocks at the point of access

can carry lasting consequences. In science, [Azoulay et al. \(2014\)](#) document a Matthew effect in which early publication success cascades into future productivity gains, and [Wang et al. \(2019\)](#) report that scientists who narrowly miss early-career grant funding subsequently outperform those who narrowly succeed. [Azoulay et al. \(2010\)](#) show that the death of a superstar scientist reduces the productivity of their collaborators, so access to high-quality peers shapes output. College access at the admission threshold produces long-run earnings gains ([Zimmerman, 2014](#); [Hoekstra, 2009](#); [Goodman et al., 2017](#)). In sports, [Berger and Pope \(2011\)](#) show that NBA teams trailing at halftime are more likely to win, giving evidence on motivational responses to adversity. This paper contributes to this literature on access shocks by combining literal randomization with the ability to separately measure access and ability responses, a distinction that most existing studies of opportunity provision cannot make.

A third strand uses professional sports as laboratories for testing economic theories. [Gauriot and Page \(2019\)](#) use close tennis matches to show that the ranking system overrewards luck relative to skill. [González-Díaz et al. \(2012\)](#) study whether tennis players perform better in high-stakes moments. [Sunde \(2009\)](#) tests tournament theory predictions about heterogeneity and incentives. [Ehrenberg and Bognanno \(1990\)](#) examines whether golf tournament prize structures have incentive effects. Most directly related, [Maity et al. \(2025\)](#) study lucky losers across professional sports and report that early-career setbacks are associated with stronger subsequent performance. Their analysis is correlational: they compare lucky losers to other qualifying losers without accounting for the selection that determines who receives a lucky loser slot. This paper provides the first causal estimates by exploiting the Grand Slam lottery mechanism.

The gap this paper fills is the absence of clean causal identification combined with dynamic career tracking. Existing studies of initial conditions use quasi-experimental variation that is typically one-shot (graduating in a recession, narrowly winning an election) and measure outcomes at a single long-run horizon. The lucky loser setting delivers literal randomization, dynamic outcomes at multiple horizons from 4 to 52 weeks, and separate

measurement of access and ability. The finding that access effects persist while ability effects are null on the ATP tour (and marginally positive on the WTA tour) is a distinction the existing literature is not positioned to make because it typically observes only one dimension of career capital.

3 Institutional Setting and Data

3.1 Institutional Background

Professional tennis is organized into a hierarchical tournament structure by the Association of Tennis Professionals (ATP, men) and the Women’s Tennis Association (WTA, women). Four Grand Slam tournaments (the Australian Open, Roland Garros, Wimbledon, and the US Open) sit at the top of the hierarchy, awarding up to 2,000 ranking points for the winner. Below them are Masters 1000 events, followed by ATP/WTA 500 and 250 events. Each tournament awards ranking points based on the round reached, with higher-tier events offering more points per round. Both tours use a rolling 52-week ranking: a player’s ranking equals the sum of points from their best results over the previous year, updated weekly. Rankings determine direct entry into tournament main draws: the top 104 players enter Grand Slam main draws directly, the top 40–56 enter Masters events, and similar thresholds apply at lower tiers. This structure creates a feedback loop central to the analysis: ranking points determine tournament access, and tournament access determines opportunities to earn ranking points.

Players ranked below the direct-entry threshold can enter a tournament’s qualifying draw, a mini-tournament held before the main event. Qualifying draws are single-elimination brackets (typically 32 or 16 players on the ATP tour, 12 or 16 on the WTA tour), with 4 to 8 qualifiers advancing to the main draw after winning two or three consecutive matches. The players who populate qualifying draws are ranked approximately 100th to 300th in the world, too low for direct entry but competitive enough to warrant a spot in the qualifying

field. These marginal players are the focus of this study. The two tours differ in ways that matter for the analysis. The WTA calendar is denser, with more tournaments per month and shorter gaps between events. WTA qualifying draws are typically smaller (often 12 players compared to 16 on the ATP side), so fewer final-round qualifying losers are generated per event.

When a player accepted into a tournament main draw withdraws (due to injury, illness, or scheduling conflicts), the vacant slot is filled by a lucky loser: a player who lost in the final round of qualifying. The selection mechanism differs by tournament tier, and this difference is central to the identification strategy. At Grand Slam tournaments, the rulebook specifies a two-track system based on withdrawal timing. When a main-draw withdrawal occurs *before* qualifying is complete, the top four ranked final-round qualifying losers enter a random draw (lottery) for the available slot; each of the four has an equal probability of selection regardless of ranking differences within the group. When a withdrawal occurs *after* qualifying ends, the highest-ranked final-round qualifying loser receives the slot without a lottery. The lottery pathway commenced after Wimbledon 2005 and provides the basis for the primary identification strategy. At non-Grand-Slam events (Masters 1000, ATP 500, ATP 250), lucky loser selection follows a strict ranking rule: the highest-ranked final-round qualifying loser receives the first available slot, the second-highest receives the next, and so on. There is no lottery component.

I supplement ranking points with Elo ratings as a measure of playing ability independent of the ranking-point system. Elo ratings update after each match based on the outcome and both players' pre-match ratings; beating a higher-rated opponent produces a larger gain than beating a lower-rated one. I compute Elo ratings from the full match history using a K -factor of 32 and an initial rating of 1,500. The distinction between ranking points (which reward participation regardless of opponent quality) and Elo ratings (which adjust for opponent quality) is central to the mechanism analysis: if LL entry improves rankings but not Elo, the improvement reflects more participation rather than improved ability.

3.2 Data Sources and Sample Construction

The primary data source is Jeff Sackmann’s Tennis Abstract database, a comprehensive public dataset of professional tennis match results covering the ATP tour from 1968 to 2024 and the WTA tour over the same period. The database includes match outcomes, player rankings, tournament details (surface, tier, draw size), and entry type (direct acceptance, qualifying, wildcard, or lucky loser). I supplement match records with weekly ranking files to construct ranking trajectories before and after each event. Elo ratings are computed from the full match history using the parameters described above.

The unit of observation is a player-episode pair (i, e) , where player i is a final-round qualifying loser at event e . The eligible pool consists of all final-round qualifying losers at Grand Slam events where at least one lucky loser slot was awarded, restricted to the top four ranked losers per event and to players with *no prior lucky loser awards of any type*. This LL-naive restriction ensures that both treated and control players enter the lottery pool without any prior main-draw access through the LL mechanism. Players at events with zero LL entries are excluded because the absence of main-draw withdrawals means no player at that event faced any probability of receiving LL treatment. Grand Slam qualifying data with lottery-based assignment are available from 2006 to 2024.

The treatment variable is defined as

$$D_{ie} = 1 \text{ if player } i \text{ entered the main draw as a lucky loser at event } e.$$

For the Grand Slam lottery design, treatment assignment is random within the top-four pool. For the non-Grand-Slam control-function design, treatment is determined by the ranking rule, and endogeneity is corrected using a control function derived from the peer component of the selection probability.

The primary estimation sample consists of first-time LL candidates among the top-four ranked final-round qualifying losers at Grand Slams from 2006 to 2024. Both treated and

control players have zero prior LL awards of any type (Grand Slam or non-Grand-Slam), so the estimated effect captures the impact of a first-ever access shock. The first-time ATP Grand Slam sample contains $N = 120$ player-episode observations (61 treated, 59 control) across 41 events with identifying variation. The first-time WTA Grand Slam sample contains $N = 82$ observations (33 treated, 49 control). The *first-time non-Grand-Slam sample* applies the same LL-naive restriction and contains $N = 2,054$ ATP observations (523 treated, 1,531 control) across 266 events with variation, and $N = 1,423$ WTA observations (378 treated, 1,045 control) across 258 events.¹ The non-Grand-Slam control function is re-estimated on the first-LL-restricted event set. The robustness section reports a marginal impact analysis that pools all LL recipients (including repeat ones) and compares treatment effects across experience levels (Section 7.1).

3.3 Lucky Loser Distribution

LL entries are not rare events in the first-LL sample. Across the 41 ATP Grand Slam events with identifying variation, 61 of 120 top-four LL-naive pool members received LL entry, a rate of 51 percent. On the WTA side, 33 of 82 pool members (40 percent) received entry across 32 events. The lower WTA rate reflects smaller qualifying draws and fewer withdrawal opportunities. In the non-Grand-Slam sample, 523 of 2,054 ATP pool members received LL entry (25 percent) and 378 of 1,423 WTA members received entry (27 percent); the lower rates reflect larger pool sizes at ATP 250/500 and WTA 250/500 events.

3.4 Variable Construction

Outcomes are organized into two categories. *Immediate event-level outcomes* measure the direct effect of the LL slot at the focal event: whether the player won at least one main-draw match, the number of main-draw matches played, and ranking points earned at the event.

¹WTA 1000 events (formerly Premier Mandatory) correspond to ATP Masters 1000; WTA 500 (formerly Premier) to ATP 500; WTA 250 (formerly International) to ATP 250.

Post-episode dynamic outcomes measure career trajectories at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks following the event. Primary dynamic outcomes include the number of main-draw entries (MainDraws_{ie}^h), the number of matches played at ATP/WT A 250+ events (Matches250_{ie}^h), and the change in ranking points ($\Delta\text{Points}_{ie}^h$). Secondary outcomes include the change in Elo rating (ΔElo_{ie}^h). The Elo outcome is the key test of ability: if LL entry improved playing skill, Elo ratings would rise.

Pre-draw state variables enter as controls:

$$\begin{aligned} Z_{ie}^{pre} = \{ & \text{Points}_{ie}/1000, (\text{Points}_{ie}/1000)^2, \\ & \text{Elo}_{ie}/100, (\text{Elo}_{ie}/100)^2, \\ & \text{SurfElo}_{ie}/100, (\text{SurfElo}_{ie}/100)^2, \text{Age}_i \} \end{aligned}$$

Pre-treatment controls include ranking points / 1000 (linear and quadratic), Elo rating / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. Prior-LL count variables are excluded because they are zero by construction for all players in the first-LL-restricted sample. Elo ratings are divided by 100 when used as covariates to improve coefficient scale (raw Elo ranges from about 1,400 to 2,200); Elo *outcomes* (change in Elo rating) remain in raw points for direct interpretability. Ranking points and Elo absorb the mechanical confound that higher-ranked LL candidates have better tournament access regardless of LL entry. Age captures career-stage effects. In pooled ATP–WT A specifications, a tour indicator is added.

Only pre-treatment variables are included as controls. Conditioning on post-treatment variables (for example, contemporaneous Elo or ranking at horizon h) would absorb part of the causal pathway. Given the sample sizes (120 observations in the first-LL ATP lottery sample, 82 in the WT A sample), I favor simpler models with fewer explanatory variables to preserve degrees of freedom.

3.5 Summary Statistics

Summary statistics for the ATP and WTA first-LL Grand Slam estimation samples are reported in Appendix Tables 10 and 11.

Players in the Grand Slam lottery pool are ranked approximately 100th to 250th in the world, with a mean Elo rating around 1,700. Ages range from 18 to 35, with a mean of approximately 24 years. The distribution is right-skewed in ranking, with a few players near the top 100 mixed with many ranked 150–250, and approximately symmetric in Elo.

The balance tests (reported in Section 7.4) show that the first-time ATP sample has a joint F -test p -value of 0.106 and the first-time WTA sample has a joint F -test p -value of 0.063. Within-event balance tests yield $p = 0.046$ (ATP) and $p = 0.516$ (WTA); the within-event test is the economically relevant specification since randomization operates within each event’s top-four pool.

The players in this sample are drawn from the same competitive margin: final-round qualifying losers ranked in the top four at their event, with no prior LL awards of any type. They are heterogeneous in age, ranking, and career stage, and the randomization at Grand Slams ensures that treatment and control groups come from the same distribution. The first-time non-Grand-Slam samples have similar characteristics but are larger (ATP: $N = 2,054$; WTA: $N = 1,423$), giving greater statistical power at the cost of relying on the control function for identification. Non-Grand-Slam summary statistics are reported in the appendix.

4 Econometric Framework and Identification

4.1 Grand Slam Identification Strategy

I restrict both the Grand Slam and non-Grand-Slam samples to *first-time* lucky loser candidates: players who have never previously received a lucky loser slot of any type (Grand

Slam or non-Grand-Slam). Both treated players ($D_{ie} = 1$, received the LL slot) and controls ($D_{ie} = 0$, eligible but not selected) are LL-naive: they have zero prior LL awards. This restriction ensures that the estimated effect captures the impact of a first-ever access shock, free of contamination from prior LL experience, and simplifies the pre-treatment control vector Z_{ie}^{pre} by eliminating prior-LL count variables that are zero by construction.

The Grand Slam lottery provides random assignment of lucky loser entry within the top-four pool. The identifying assumption is

$$E[Y_{ie}(0) \mid D_{ie} = 1, e] = E[Y_{ie}(0) \mid D_{ie} = 0, e],$$

so among the top four ranked final-round qualifying losers at a given Grand Slam, lottery-selected and unselected players have the same expected untreated potential outcomes. This holds by design: within the pool, each player has an equal probability of selection regardless of ranking, ability, or other characteristics.

The randomization operates within each event’s top-four pool, not across players globally. A player ranked 120th at the Australian Open and a player ranked 180th at Roland Garros are not comparable by virtue of the lottery alone; they are in different pools. The event fixed effect α_e absorbs all between-pool variation, so the estimated effect compares lottery winners to lottery losers within the same event.

The lottery operates only when withdrawals occur before qualifying completes. Post-qualifying withdrawals follow the ranking rule, so the sample may include some ranking-based assignments. The verified lottery subsample (Section 7.3) addresses this concern by restricting to events with at least one LL entry whose rank among qualifying losers exceeds one, a data-only criterion that identifies 35 verified ATP events and 17 verified WTA events.

With only four players per pool and a binary treatment, finite-sample imbalance is expected even under true randomization. In the first-time ATP Grand Slam sample ($N = 120$), the joint F -test yields $p = 0.106$; on the WTA side ($N = 82$), $p = 0.063$. Neither rejects at

the 5% level, though both are closer to marginal than in the full sample, reflecting the smaller first-LL-restricted pool. To improve precision and absorb residual imbalance, I include pre-treatment controls in all dynamic specifications; I also report specifications without controls as a complement.

4.2 Immediate Effects

The event-level model estimates the immediate effect of receiving the LL slot at the focal Grand Slam:

$$Y_{ie}^{imm} = \alpha_e + \delta D_{ie} + \varepsilon_{ie}, \quad (1)$$

where Y_{ie}^{imm} is one of three event-level outcomes: an indicator for winning at least one main-draw match, the number of main-draw matches played, and ranking points earned at the event. The event fixed effect α_e is a Grand Slam $\times$ Year indicator. The coefficient δ measures the size of the initial treatment shock: the immediate dose of ranking points and competitive experience that the LL slot provides. This quantity sets the scale for interpreting the dynamic effects that follow. Control outcomes are mechanically zero because unselected players do not enter the main draw.

4.3 Post-Episode Dynamics

The dynamic model traces career outcomes at horizons $h \in \{4, 8, 12, 26, 52\}$ weeks:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}, \quad (2)$$

where $Y_{ie,h}$ is the outcome for player i from event e measured at horizon h , α_e is an event fixed effect (Grand Slam $\times$ Year), α_h is a horizon fixed effect absorbing level differences across time horizons, Z_{ie}^{pre} is the vector of pre-treatment controls (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and player age) with horizon-invariant coefficients Γ , and $\varepsilon_{ie,h}$ is the error term. Standard

errors are clustered at the player level because players may receive repeated LL opportunities across events, inducing within-player dependence across episodes and horizons; event-level clustering is unnecessary because event fixed effects already absorb between-event variance. The coefficients on Z^{pre} do not vary by horizon because these controls absorb pre-treatment differences that are constant across the post-event trajectory. The horizon fixed effects are important for count outcomes (main draws entered, matches played) where level differences across horizons are substantial; adding α_h raises the R^2 from 0.35 to 0.53 for main-draw entries.

The coefficient β_h identifies the causal effect of lottery selection on the outcome at horizon h . The sequence $\{\beta_4, \beta_8, \beta_{12}, \beta_{26}, \beta_{52}\}$ traces the divergence between treated and control trajectories after the random shock. Immediate effects ($h = 0$) are captured by the event-level model in equation (1) and are excluded from the dynamic specification to avoid double-counting.

The estimand β_h is the average treatment effect on the treated (ATT) at horizon h , averaged across events with weights proportional to the number of first-time LL candidates in the top-four pool. The event fixed effects ensure that comparisons are within-pool, and the stacked specification pools across horizons to improve precision. Because the lottery assigns treatment with equal probability within each pool, the ATT and ATE coincide within the experimental population.

A player may appear in the lottery pool at multiple events. To avoid contamination from subsequent treatments, I censor each player-episode at the date of the player’s next final-qualifying-round loss at an LL-granting event of the same type:

$$h^* = \min(12 \text{ months}, \text{time to next same-type qualifying loss at an LL-granting event}).$$

For Grand Slam episodes, this censors at the next Grand Slam qualifying loss where at least one LL slot was awarded; for non-Grand-Slam episodes, censoring occurs at the next

non-Grand-Slam qualifying loss at an LL-granting event. Within this window, the only relevant treatment is D_{ie} from the originating event, so overlapping treatment spells do not contaminate the estimate. A player who appears in the LL candidate pool at multiple events generates multiple episodes, potentially with different treatment statuses. As a robustness check, I restrict the sample to each player’s first Grand Slam LL opportunity (Section 7.1).

The dynamic model’s 4-week horizon measures outcomes *after* the focal event. Because the focal event itself occupies the first 1–2 weeks of the post-treatment window, treated players may enter fewer *additional* main draws during this period than controls who are free to enter other events; the immediate-effects model (equation 1) captures the focal event’s outcomes separately. The negative 4-week main-draws coefficient in the non-GS sample reflects this mechanical timing effect; by 12 weeks the treatment effect on main draws becomes positive as ranking gains open new entry opportunities.²

4.4 Heterogeneous and Treatment-Dose Effects

I estimate heterogeneous treatment effects by augmenting equation (2) with interactions between the treatment indicator D_{ie} and category dummies. For a categorical moderator with K groups (one base category omitted), the estimating equation is

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \sum_{k=2}^K \delta_{k,h} D_{ie} \times \mathbf{1}[G_{ie} = k] + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}, \quad (3)$$

where β_h captures the base-category treatment effect at horizon h and $\delta_{k,h}$ captures the differential effect for category k relative to the base. Because β_h is already indexed by horizon in equation (2), this specification inherits the horizon-specific treatment structure without additional horizon interactions. I examine three dimensions: pre-treatment ranking (above versus below the tour-specific median, with below-median as base), age (below versus above 24, with younger as base), and prior LL experience (none versus any, with no prior

²This timing pattern is specific to count outcomes (main draws entered, matches played) and does not affect ranking-point or Elo outcomes, which cumulate over the horizon window inclusive of the focal event.

experience as base).

To characterize how treatment effects vary with dose, I interact D_{ie} with the number of main-draw matches won at the focal event, entered continuously. The dose variable is endogenous: players who win more matches are better or luckier on the day, so these estimates are descriptive rather than causal. They answer a different question than the ITT: conditional on receiving the LL slot, how much does the career impact depend on what the player does with it? Standard errors are clustered at the player level throughout.

4.5 Non-Grand-Slam Control-Function Strategy

At non-Grand-Slam events, LL slots go to the highest-ranked final-round qualifying losers. I use a control-function approach to account for endogenous LL assignment, exploiting the peer component of the selection probability computed from the combinatorial structure of qualifying match pairings.

The selection probability P_i^{LL} is the probability that player i 's rank among final-round qualifying losers falls within the top c slots (where c is the number of LL positions available), conditional on player i having lost:

$$P_i^{LL} = \Pr(\text{rank among losers} \leq c \mid i \text{ loses}).$$

This probability depends on the qualifying match outcomes of *other* players, specifically on the probability that enough higher-ranked qualifiers also lose their matches. Construction proceeds in two steps. First, I estimate a weighted logit model of match outcomes on qualifying-round matches at non-Grand-Slam LL-granting events, separately for each tour (ATP: $N = 9,775$; WTA: $N = 10,199$ qualifying matches). The match-level covariates X_{ijm} include ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed head-to-head win proportion, head-to-head encounter count, and age difference, along with

event-level surface indicators (clay and grass). The logit is estimated by weighted MLE, with weights equal to the total points in play (sum of service points) in each match; qualifying round number serves as a fallback weight when service point data are unavailable. Estimating on all qualifying rounds, not just the final round, sharpens the parameter estimates by exploiting the full set of qualifying matches. The same logit specification is used for the selection probability, the tournament performance model, and the treatment-dose analysis, so the covariate structure is unified across applications. This yields predicted match-winning probabilities $\hat{P}(a_m \rightarrow b_m \mid X, Z)$ for each final-round qualifying match m between players a_m and b_m . Second, given these predicted probabilities and the $N_e - 1$ other final-round qualifying matches (excluding player i 's match, since i is conditioned on having lost), I enumerate all 2^{N_e-1} possible outcome combinations. Each combination has probability equal to the product of the individual match outcome probabilities from the logit, invoking conditional independence of match outcomes given observables. Conditional independence is plausible because each final-round qualifying match involves a distinct pair of players with independent on-day shocks, and the logit already conditions on ranking, Elo, surface Elo, head-to-head history, age, and surface; residual correlation across same-event matches would have to arise from unobserved event-day conditions shared across disjoint player pairs, which is a priori weak. For each combination, the set of losers is determined and their rankings compared; P_i^{LL} is the sum of probabilities of all combinations in which i 's rank falls within the top c among losers. Player i 's own loss probability does not enter P_i^{LL} ; the enumeration uses only final-round matches, since the LL selection mechanism conditions on the final qualifying-round structure. The resulting P_i^{LL} is systematically lower than a naive Bernoulli convolution that ignores the draw structure (mean 0.279 vs. 0.506 across the non-GS sample), because the exact enumeration correctly conditions on which players are paired against each other. The key identifying assumption is that, conditional on player i 's characteristics and the fact that i lost, the qualifying-match outcomes of other players affect i 's future career only through the lucky-loser channel.

For all non-Grand-Slam specifications, I account for endogenous LL assignment using a control-function approach. The generalized residual $\hat{v}_{ie}$ is computed analytically from P_i^{LL} :

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}},$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF. This residual captures how far player i 's actual LL status deviates from what P_i^{LL} would predict under normality. It enters the outcome equation as an additional regressor:

$$Y_{ie,h} = \alpha_e + \alpha_h + \beta_h D_{ie} + \rho \hat{v}_{ie} + \Gamma Z_{ie}^{pre} + \varepsilon_{ie,h}. \quad (4)$$

The coefficient $\hat{\rho}$ tests for endogeneity of LL entry: $\hat{\rho} \neq 0$ indicates that LL assignment is correlated with unobservables and the control-function correction is needed. The ATP non-Grand-Slam sample contains $N = 2,054$ observations (523 treated); the WTA sample contains $N = 1,423$ (378 treated). Standard errors are clustered at the player level.

For the match-level tournament performance model at non-Grand-Slam events (Section 6.2), the same analytically computed generalized residual enters the outcome logit:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta \cdot D_{ie} + \rho \cdot \hat{v}_{ie}), \quad (5)$$

where X_{ijm} includes ranking points difference / 1000 (linear and quadratic), Elo difference / 100 (linear and quadratic), surface Elo difference / 100 (linear and quadratic), Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and surface indicators (clay and grass): the same logit specification used for the selection probability. Z_{ie}^{pre} includes ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and player age. In the dose specification, $\hat{v}_{ie}$ is interacted with the centered matches-won variable to allow the endogeneity correction to vary with treatment dose, because players who win more matches at the LL event may have different unobserved

ability profiles. Standard errors are clustered at the player level; the same clustering applies to the delta specification and preserves within-player dependence across events and matches.

5 Results

Grand Slam results exploit the lottery for identification and are presented separately for the ATP and WTA tours. Non-Grand-Slam control-function results are summarized in Section 5.4 and reported in full in the appendix.

5.1 Immediate Effects

Table 1 characterizes the treatment dose at the focal event. Because control players do not enter the main draw, these estimates describe the magnitude of the treatment rather than a causal effect against a counterfactual.

On the ATP tour ($N = 120$), treated players play additional main-draw matches and earn ranking points proportional to the round reached; the WTA pattern ($N = 82$) is similar. A Grand Slam first-round loss earns 10 ranking points and a second-round loss earns 45 points, so even a single main-draw win delivers an economically meaningful shock to the player’s rolling 52-week ranking-point stock.

5.2 Post-Episode Dynamics

Tables 2 and 3 report estimates from a stacked specification with horizon-specific treatment effects estimated jointly. The stacked model pools all five horizons into a single regression with horizon-specific LL coefficients and horizon fixed effects, improving efficiency by borrowing information across horizons while allowing the treatment effect to vary freely at each horizon.

On the ATP side, the sample comprises 120 first-time player-episodes contributing approximately 600 stacked observations across five horizons. The lottery raises ranking points

Table 1: Grand Slam Lottery: Immediate Event-Level Effects

Outcome	<i>t</i> -test	Event FE	
	LL Mean	$\hat{\beta}$	(SE)
<i>ATP</i>			
Main-draw match win	0.34***	0.39***	(0.09)
Main-draw matches played	1.46***	1.64***	(0.13)
Ranking points at event	28.85***	36.35***	(5.50)
<i>N</i>	61 / 59	120	
<i>WTA</i>			
Main-draw match win	0.36***	0.33***	(0.09)
Main-draw matches played	1.42***	1.38***	(0.12)
Ranking points at event	25.45***	23.93***	(4.35)
<i>N</i>	33 / 49	82	
<i>Pooled</i>			
Main-draw match win	0.35***	0.38***	(0.06)
Main-draw matches played	1.45***	1.53***	(0.09)
Ranking points at event	27.66***	31.32***	(3.97)
<i>N</i>	94 / 108	202	

Notes. First-time Grand Slam LL candidates: top-four ranked final-round qualifying losers at events with at least one LL slot, 2006–2024. ATP: $N = 120$; WTA: $N = 82$. Event FE columns report OLS with Grand Slam $\times$ Year fixed effects. Control outcomes are mechanically zero because unselected players do not enter the main draw. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

by 72.5 at 4 weeks, 72.1 at 8 weeks, and 82.8 at 12 weeks (each significant at conventional levels), before peaking and starting to fade: the effect is 72.6 at 26 weeks (clustered $p = 0.047$; Fisher $p = 0.164$) and becomes indistinguishable from zero at 52 weeks. The 4-, 8-, and 12-week estimates survive Fisher randomization inference ($p_{Fisher} < 0.001$, $p_{Fisher} = 0.002$, $p_{Fisher} = 0.028$), so at those horizons the finding does not rely on the asymptotic approximation. The 26-week estimate does not survive Fisher inference and is the horizon at which small-sample noise begins to bite; it is the least robust of the significant effects. Benjamini-Hochberg q -values across the 80-test family of primary outcomes are reported in Appendix Table 25: the ATP GS 4-, 8-, and 12-week ranking-point effects survive joint BH correction at $q < 0.05$; the 26-week effect does not ($q = 0.119$), and 23 of the 80 tests survive overall. Elo rating changes are null at every horizon, so the ranking-point gains come from more matches and rounds played rather than from measured ability improvement. The trajectory is consistent with the initial point shock being real but temporary: it accumulates through 12–26 weeks as the ranking boost opens further tournaments, then dissipates as the original points age out of the rolling window.

On the WTA side, the 82-episode sample delivers approximately 410 stacked observations. Ranking points rise by 55.0 at 8 weeks ($p = 0.017$); estimates at other horizons are positive but imprecise. The Elo pattern differs from the ATP tour: WTA first-timers show Elo gains at 8 weeks (+15.0, $p = 0.082$) and at 52 weeks (+36.8, $p = 0.055$). These estimates should be read against the borderline joint balance ($p = 0.063$) and the smaller sample. The Elo channel on the WTA is explored further in Section 6.

Figures 1 and 2 plot mean cumulative ranking points earned by LL-selected and unselected players from 12 weeks before to 52 weeks after each Grand Slam. Pre-event trajectories on both tours are parallel, so pre-treatment divergence is not detectable. After the event, ATP LL recipients accumulate approximately 72 additional ranking points by 4 weeks and 83 by 12 weeks; the advantage then narrows as the rolling window absorbs the initial gains. WTA LL recipients diverge at 8 weeks by 55 points, with controls catching up more quickly

Table 2: ATP Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	72.5*** (26.7)	72.1*** (26.6)	82.8*** (26.1)	72.6** (36.2)	23.5 (59.1)
Elo Δ	-4.4 (9.1)	0.2 (10.9)	-4.8 (10.3)	4.7 (13.5)	-9.7 (21.0)
Main Draws	-0.2 (0.5)	0.1 (0.5)	0.1 (0.5)	0.7 (0.5)	0.8 (1.5)
Matches 250+	0.3 (1.0)	0.7 (1.0)	0.9 (0.9)	1.5 (1.1)	1.3 (2.8)
Win %	0.04 (0.05)	-0.01 (0.03)	-0.03 (0.03)	-0.03 (0.03)	-0.03 (0.03)

Notes. Stacked specification: 120 first-time player-episodes contributing approximately 600 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls (ranking points linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and age). Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3: WTA Grand Slam: Post-Episode Dynamic Effects (Stacked Model)

	4w	8w	12w	26w	52w
Ranking Pts Δ	40.0 (28.5)	55.0** (22.5)	27.8 (20.1)	7.5 (42.8)	26.5 (86.4)
Elo Δ	13.3* (7.6)	15.0* (8.5)	11.8 (9.1)	21.6* (12.7)	36.8* (18.9)
Main Draws	-0.1 (0.5)	0.3 (0.5)	0.3 (0.6)	0.7 (0.7)	1.8 (1.5)
Matches 250+	-0.5 (1.0)	0.8 (0.7)	0.6 (0.8)	1.9* (1.0)	3.2 (2.5)
Win %	0.03 (0.05)	0.05* (0.03)	0.05* (0.03)	0.04 (0.03)	0.05 (0.03)

Notes. Stacked specification: 82 first-time player-episodes contributing approximately 410 stacked observations across five horizons (4, 8, 12, 26, 52 weeks). Horizon-specific LL treatment effects estimated jointly with horizon fixed effects, Grand Slam $\times$ Year fixed effects, and pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

than on the ATP tour, a pattern consistent with the denser WTA calendar.

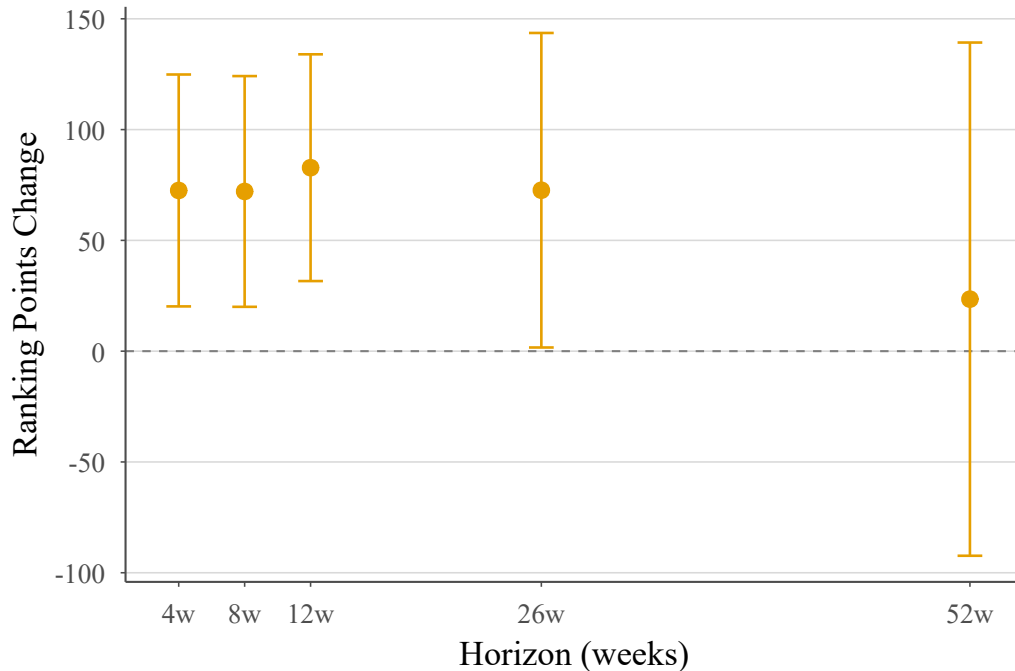


Figure 1: ATP Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point is the estimated treatment effect at the indicated horizon. Vertical dashed line marks the event date. Shaded regions are 95 percent confidence intervals. First-time ATP Grand Slam sample ($N = 120$).

5.3 Heterogeneous Effects

The first-time GS sample is small ($N = 120$ ATP, $N = 82$ WTA), limiting statistical power for subgroup analysis. Table 4 reports heterogeneity for the ATP GS sample. Effects are stronger and more persistent for older players (age ≥ 24): ranking-point gains of 154.9 at 26 weeks ($p < 0.001$) and 186.4 at 52 weeks ($p < 0.01$). This pattern is consistent with older players at the qualifying margin having more tournament experience to convert a ranking boost into sustained access, even with a shorter remaining career runway. Other subgroup contrasts, by pre-episode ranking and by surface, are imprecisely estimated in the small ATP sample.

The non-GS samples provide substantially more power for heterogeneity analysis ($N =$

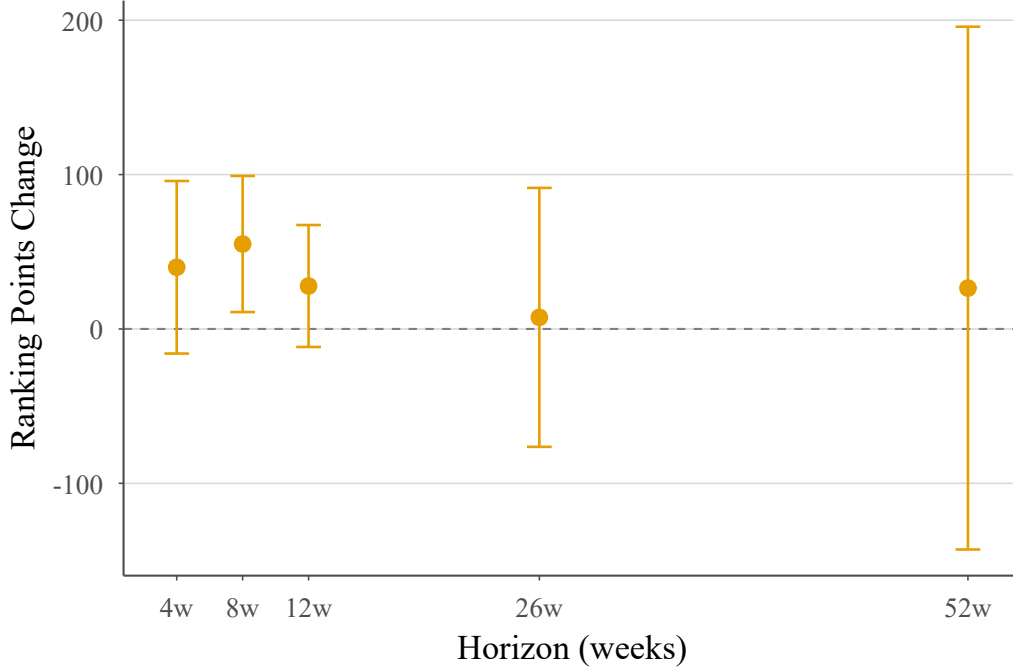


Figure 2: WTA Event Study: Ranking Trajectories for Selected and Unselected Players

Notes. Each point is the estimated treatment effect at the indicated horizon. First-time WTA Grand Slam sample ($N = 82$).

2,054 ATP, $N = 1,423$ WTA); results are reported in Appendix Tables 14 and 15.

5.4 Non-Grand-Slam Extension

The non-Grand-Slam control-function estimates provide corroborating evidence from substantially larger samples. On the ATP tour ($N = 2,054$), ranking-point gains are 20.6 at 4 weeks ($p = 0.041$) and 44.6 at 26 weeks ($p = 0.002$). Elo effects emerge at medium horizons: +12.8 at 8 weeks ($p = 0.018$) and +30.8 at 26 weeks ($p = 0.016$), a pattern that differs from the GS null and may reflect either the larger sample’s power to detect moderate ability effects or a difference in the non-GS competitive environment.

On the WTA tour ($N = 1,423$), ranking-point gains are 23.2 at 12 weeks ($p = 0.057$), with Elo gains of 10.5 at 12 weeks ($p = 0.011$). The WTA non-GS Elo finding parallels the marginal WTA GS Elo result and points toward genuine, if modest, ability gains, plausibly linked to greater competitive parity on the women’s tour.

Table 4: ATP Grand Slam: Heterogeneous Effects

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	71.5** (33.5)	76.5** (30.7)	91.4*** (28.1)	79.6* (44.5)	26.1 (75.4)
$\times$ High	0.6 (42.9)	-10.3 (39.7)	-18.7 (39.8)	-15.6 (47.1)	-6.1 (73.4)
Main Draws	-0.7 (0.6)	-0.4 (0.6)	-0.2 (0.6)	0.8 (0.6)	1.3 (1.7)
$\times$ High	1.2 (0.8)	0.9 (0.8)	0.8 (0.8)	-0.3 (0.9)	-1.0 (2.1)
Elo Δ	-8.8 (9.5)	0.6 (10.5)	-4.3 (10.7)	22.3 (16.5)	-3.7 (25.0)
$\times$ High	8.1 (14.5)	-1.4 (15.2)	-1.5 (16.2)	-34.7* (19.1)	-12.2 (25.9)
<i>Age</i>					
Ranking Pts Δ	76.1** (32.8)	56.6 (34.3)	71.6** (33.8)	6.1 (40.2)	-49.0 (68.3)
$\times$ High	-9.9 (37.3)	35.8 (38.4)	27.8 (39.2)	154.9*** (47.2)	186.4** (74.3)
Main Draws	-0.2 (0.6)	-0.3 (0.6)	-0.3 (0.6)	-0.4 (0.6)	-1.3 (1.7)
$\times$ High	0.1 (0.7)	0.9 (0.7)	1.1* (0.6)	2.5*** (0.8)	5.2** (2.1)
Elo Δ	-9.7 (11.5)	-7.1 (13.0)	-14.7 (12.9)	-13.6 (15.4)	-19.5 (24.1)
$\times$ High	13.9 (15.2)	18.6 (15.7)	24.5 (17.3)	45.7** (19.2)	23.7 (25.5)

Notes. Stacked specification with horizon-specific LL treatment effects estimated by subgroup. First-time ATP Grand Slam sample ($N = 120$), 2006–2024. Low/High rank points split at the tour-specific median. Younger/Older split at age 24. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The non-GS estimates align with the GS lottery estimates in both direction and magnitude, supporting the validity of the control-function approach. Full non-Grand-Slam dynamic tables are reported in Appendix Tables 12 and 13; heterogeneity results are in Appendix Tables 14 and 15.

The dose-response analysis examines whether treatment effects vary with the number of main-draw matches won at the focal event. The initial LL slot provides a ranking-point benefit that is amplified when the player wins matches in the main draw. Dose-response results are reported in the appendix.

6 Mechanisms

The results establish that first-time lucky loser entry causes gains in ranking points and tournament access. This section decomposes the effect into two channels, access and ability, and examines match-level performance conditional on opponent strength.

6.1 Access versus Ability

The career effects of LL entry operate primarily through tournament access rather than improved competitive skill. On the ATP tour, ranking points rise significantly through 26 weeks while Elo ratings are null at every horizon. Because Elo adjusts for opponent quality, the null Elo result indicates that treated players do not perform better or worse per match once opponent strength is held constant. The ranking-point gains come entirely from additional tournament participation: LL entry provides a main-draw appearance that generates ranking points, which improve the player’s position in the rolling 52-week ranking, which crosses direct-entry thresholds for subsequent lower-tier events. The evidence aligns with this access cascade: treated players enter more main draws and play more matches at 250+ events over 26 weeks, and their quality-adjusted match performance is unchanged.

The WTA tour presents a nuance. Elo gains at 8 weeks (+15.0, $p = 0.082$) and 52 weeks

(+36.8, $p = 0.055$) are consistent with WTA LL recipients experiencing modest ability improvements. The same pattern appears in the non-GS WTA results, where Elo gains reach significance at 12 weeks (+10.5, $p = 0.011$). Greater competitive parity on the women’s tour would give marginal players more room to accumulate ability gains through main-draw experience against stronger opponents. The match-level tournament performance model (Section 6.2) does not detect a significant WTA effect on win probability conditional on opponent strength, however, so the ability interpretation carries only limited weight. The tension between Elo gains in the stacked model and the null match-level win probability estimate ($\hat{\delta} = +0.146$, $p = 0.29$; player-clustered) can plausibly reflect Elo’s sensitivity to the schedule of opponents faced. If treated WTA players enter more tournaments with weaker fields (lower-tier events where ranking gains provide access), Elo can improve even without a genuine ability change, because the Elo system rewards wins against any opponent. The match-level model conditions on actual opponent strength and provides a stricter test of ability improvement.

The access cascade is self-limiting. As the 52-week rolling window progresses, the points from the original LL event constitute a shrinking fraction of the player’s total. Unless downstream tournaments generate enough additional points to replace the original ones, the ranking converges back toward the level determined by sustained match performance. The dynamic results align with this: ATP ranking-point gains peak at 12 weeks and fade toward zero at 52 weeks.

6.2 Match-Level Performance

To test directly whether LL entry affects competitive performance itself, I estimate a match-level logit of win probability conditional on opponent strength and tournament context (see Appendix A.1). At Grand Slams, the lottery provides exogenous variation; at non-GS events, a control function corrects for endogenous selection.

First-time ATP GS LL recipients have $\hat{\delta} = -0.141$ (SE = 0.103, $p = 0.17$; player-

clustered), a point estimate consistent with overseeding when novice entrants face opponents above their ability level, though not significant at conventional levels. WTA GS recipients have $\hat{\delta} = +0.146$ (SE = 0.138, $p = 0.29$), also not statistically distinguishable from zero. Non-GS ATP has $\hat{\delta} = +0.609$ ($p < 0.001$), reflecting the control-function correction in a setting where treated players are positively selected on ability. The GS point estimates line up with the direction the overseeding channel predicts (negative for ATP, positive for WTA), but the small first-time GS samples ($N = 120$ ATP, 82 WTA) do not deliver enough power for individually significant tests. Full results including dose and horizon interactions are reported in Appendix Table 16.

The event study figures (Figures 1 and 2) provide a visual summary: ranking-point trajectories diverge sharply after the LL event while pre-event trends are parallel, so the treatment effect is concentrated in post-event access gains rather than pre-existing ability differences.

6.3 Economic Magnitude: Prize Money

Ranking points are an imperfect proxy for the economic stakes. To ground the results in dollar terms, I compute the implied prize-money differential using publicly available tournament prize tables. At Grand Slams, a first-time LL recipient earns \$100,000–\$400,000 USD depending on how far they advance in the main draw, versus a \$40,000 qualifying-round stipend for controls. Combined with the estimated access gains over the 26 weeks following the focal event (evaluated at an average forward-looking first-round prize of \$11,000 per additional main draw), the total prize-money effect of a first-time LL entry is approximately \$101,000 on the ATP GS sample and \$94,000 on the WTA GS sample (Appendix Table 26). For non-GS first-time LL recipients, the total dollar effect is smaller but economically meaningful: \$28,000 on the ATP tour and \$16,000 on the WTA tour, driven more heavily by the dynamic access cascade than by the immediate event prize. These magnitudes are substantial against qualifying-level players’ annual earnings, which typically fall in the \$40,000–\$80,000 range.

Under a $\pm 20\%$ sensitivity band on the per-entry prize assumption (\$8,800–\$13,200 instead of \$11,000), the ATP GS total effect ranges from approximately \$99,000 to \$103,000 and the non-GS ATP effect from approximately \$24,000 to \$31,000; the qualitative conclusion (dollar effects comparable to a qualifier’s annual earnings) is insensitive to reasonable variation in the per-entry assumption. The first break therefore carries real financial consequences even as its ranking-point effects fade within a year.

6.4 Reconciling with the Existing Literature

[Maity et al. \(2025\)](#) report positive correlations between lucky loser entry and career outcomes. The match-level tournament performance model, which exploits the lottery for identification, does not detect an improvement in match-level win probability on the ATP tour and reports only a marginal positive effect on the WTA tour. The causal estimates here indicate that the correlational finding in [Maity et al. \(2025\)](#) likely reflects the additional tournament participation that LL entry generates rather than an improvement in competitive performance itself.

The initial conditions literature reports persistent effects of early-career shocks ([Oyer, 2006](#); [Oreopoulos et al., 2012](#)). The present results align on the access dimension: LL entry causes persistent changes in the number and type of tournaments a player enters; the ability dimension is largely null. Market transparency is the plausible moderator: in professional tennis, where performance is measured match by match and rankings update weekly, ability reasserts itself within a year. The access channel alone accounts for the ranking-point gains and tournament-entry effects documented in [Section 5](#).

7 Robustness

7.1 Marginal Impact by LL Experience

The main analysis restricts to first-time LL candidates. A natural question is whether subsequent LL opportunities have similar effects, or whether the marginal impact diminishes with experience. To test this, I return to the full (unrestricted) estimation sample and stratify by the number of prior LL awards: $n = 0$ (first LL, matching the main results), $n = 1$ (second LL), and $n \geq 2$ (third or more). Within each experience group, I compare treated players ($D_{ie} = 1$) to controls ($D_{ie} = 0$) who have the same number of prior LL awards, using the full pre-treatment control specification.

Tables 5 and 6 report the results. First-time LL effects ($n = 0$) are the largest and most significant across both tours and both identification strategies, replicating the main-text findings. Second-time GS effects ($n = 1$) should be interpreted with caution: the GS $n = 1$ cells are small (ATP $N = 48$, WTA $N = 31$), event fixed effects absorb substantial variation, and the resulting point estimates are noisy and sometimes larger in absolute value than the physically possible treatment dose. The non-GS $n = 1$ cells are much larger (ATP $N = 692$, WTA $N = 601$) and produce stable estimates: the second-LL access effect is positive but markedly smaller than the first-LL effect. Third-plus effects ($n \geq 2$) in the non-GS sample are close to zero for access outcomes on the ATP tour and significantly negative for ranking points on the WTA tour at longer horizons. Formal Wald tests for the equality of $n = 0$ and $n = 1$ effects reject equality at conventional levels for non-GS tournament access outcomes (ATP: $F = 3.87$, $p = 0.002$; WTA: $F = 3.05$, $p = 0.009$). These non-GS results, estimated on larger and more reliable subsamples, support the first-break interpretation: the initial access shock is the pivotal one, and subsequent opportunities provide diminishing marginal returns.

Table 5: Marginal Impact by LL Experience: Grand Slam Samples

Tour	Experience	4w	8w	12w	26w	52w
ATP	First LL ($n = 0$)	72.5***	72.1***	82.8***	72.6**	23.5
		(26.7)	(26.6)	(26.1)	(36.2)	(59.1)
		-4.4	0.2	-4.8	4.7	-9.7
		(9.1)	(10.9)	(10.3)	(13.5)	(21.0)
		-0.2	0.1	0.1	0.7	0.8
		(0.5)	(0.5)	(0.5)	(0.5)	(1.5)
ATP	Second LL ($n = 1$)	0.3	0.7	0.9	1.5	1.3
		(1.0)	(1.0)	(0.9)	(1.1)	(2.8)
		-220.9***	-215.3***	-195.6***	-153.7**	-202.1**
		(48.5)	(44.9)	(49.0)	(64.1)	(98.6)
		-97.9***	-89.9***	-96.5***	-83.7***	-98.6***
		(17.6)	(18.3)	(21.3)	(23.0)	(26.7)
ATP	Third+ LL ($n \geq 2$)	-1.6**	-1.7***	-1.2**	0.6	1.3
		(0.6)	(0.5)	(0.5)	(0.6)	(1.9)
		-9.5***	-9.3***	-8.4***	-4.5***	-2.9
		(1.4)	(1.1)	(1.2)	(1.5)	(3.9)
		-30.7	-38.3	-40.4	4.3	-14.8
		(40.6)	(35.8)	(35.7)	(41.6)	(62.6)
ATP	Third+ LL ($n \geq 2$)	-8.3	-15.0	-18.6	-13.4	-26.5
		(15.2)	(13.6)	(14.3)	(20.5)	(20.1)
		0.4	0.7	0.8	0.5	0.7
		(0.8)	(0.7)	(0.7)	(0.8)	(1.7)
		0.9	1.0	0.9	0.7	1.8
		(1.4)	(1.2)	(1.2)	(1.3)	(2.6)
WTA	First LL ($n = 0$)	40.0	55.0**	27.8	7.5	26.5
		(28.5)	(22.5)	(20.1)	(42.8)	(86.4)
		13.3*	15.0*	11.8	21.6*	36.8*
		(7.6)	(8.5)	(9.1)	(12.7)	(18.9)
		-0.1	0.3	0.3	0.7	1.8
		(0.5)	(0.5)	(0.6)	(0.7)	(1.5)
WTA	Second LL ($n = 1$)	-0.5	0.8	0.6	1.9*	3.2
		(1.0)	(0.7)	(0.8)	(1.0)	(2.5)
		54.6	33.0	-82.8**	-154.7***	-181.3*
		(61.9)	(53.2)	(38.9)	(49.4)	(104.2)
		-6.1	-16.8	-17.9	-16.3	-52.4
		(10.0)	(10.0)	(14.8)	(18.8)	(41.5)
WTA	Second LL ($n = 1$)	-0.6	-0.2	-0.5	-0.7	-2.4
		(1.2)	(1.2)	(1.1)	(1.3)	(2.2)
		-1.0	-0.4	-0.9	-1.7	-6.0
		(2.1)	(2.0)	(1.9)	(2.1)	(3.9)

Notes. Full (unrestricted) Grand Slam sample stratified by number of prior LL awards (n). Within each group, treated = received LL, control = eligible but not selected, both with the same number of prior awards. Event FE specification with full pre-treatment controls (including prior-LL count variables). Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 6: Marginal Impact by LL Experience: Non-Grand-Slam Samples

Tour	Experience	4w	8w	12w	26w	52w
ATP	First LL ($n = 0$)	20.1**	19.7*	13.5	44.5***	37.7
		(10.2)	(10.4)	(10.3)	(14.5)	(30.4)
		6.8	11.1*	10.0*	28.0**	30.4
		(6.3)	(5.7)	(6.0)	(12.7)	(20.5)
		-1.4***	-0.9***	-0.3*	1.7***	5.2***
		(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
		-2.6***	-1.7***	-0.7**	3.0***	9.0***
		(0.4)	(0.4)	(0.3)	(0.5)	(1.2)
		6.1	11.4	2.6	19.6	-1.1
		(24.1)	(26.1)	(28.1)	(34.7)	(53.4)
ATP	Second LL ($n = 1$)	-3.5	-1.5	-1.5	-6.2	-10.9
		(7.9)	(8.6)	(9.1)	(10.4)	(14.3)
		-1.3***	-0.8**	-0.2	1.5***	4.2***
		(0.4)	(0.4)	(0.4)	(0.5)	(1.0)
		-2.4***	-1.4*	-0.6	2.7***	7.4***
		(0.8)	(0.8)	(0.8)	(1.0)	(2.0)
		11.6	11.6	15.5	19.1	-25.5
		(22.8)	(22.7)	(22.9)	(25.5)	(43.2)
		4.2	2.8	3.7	-2.0	-2.3
		(7.1)	(7.4)	(7.8)	(9.4)	(13.3)
ATP	Third+ LL ($n \geq 2$)	-0.9**	-0.4	-0.1	1.3***	3.2***
		(0.4)	(0.3)	(0.3)	(0.4)	(1.0)
		-1.4*	-0.6	0.0	2.4***	5.9***
		(0.7)	(0.7)	(0.7)	(0.9)	(1.9)
WTA	First LL ($n = 0$)	17.8	18.8*	23.0*	20.5	59.2
		(11.3)	(11.0)	(12.2)	(17.9)	(39.6)
		6.5*	7.3*	10.8***	6.4	5.7
		(3.6)	(3.8)	(4.1)	(5.8)	(8.1)
		-1.1***	-0.7***	-0.3**	0.9***	2.7***
		(0.2)	(0.1)	(0.1)	(0.2)	(0.5)
		-2.1***	-1.4***	-0.8***	1.6***	5.6***
		(0.3)	(0.3)	(0.3)	(0.4)	(1.1)
		36.6	29.6	34.4	20.9	-18.0
		(27.9)	(29.3)	(29.3)	(36.4)	(65.0)
WTA	Second LL ($n = 1$)	8.0	5.2	6.2	4.9	9.4
		(6.5)	(6.9)	(7.4)	(9.5)	(11.8)
		-1.0***	-0.6	-0.1	1.4***	3.4***
		(0.4)	(0.4)	(0.4)	(0.5)	(0.9)
		-1.6**	-1.0	-0.0	2.6***	6.2***
		(0.7)	(0.7)	(0.7)	(0.9)	(1.8)
		17.2	-1.5	-17.1	-81.4**	-159.6**
		(26.3)	(29.0)	(29.8)	(38.9)	(72.1)
		3.3	1.4	3.6	-1.4	-24.9*
		(6.4)	(7.1)	(6.9)	(10.0)	(15.0)
WTA	Third+ LL ($n \geq 2$)	-0.4	-0.0	0.4	1.1***	2.4**
		(0.3)	(0.3)	(0.3)	(0.4)	(1.1)
		-1.0	-0.6	-0.4	0.5	2.7
		(0.7)	(0.7)	(0.7)	(1.0)	(2.1)

Notes. Full (unrestricted) non-Grand-Slam sample stratified by number of prior LL awards (n). Control function specification with $\hat{v}_{ie} \times$ horizon terms and full pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 Mechanical Decomposition of the First-Break Claim

The claim that first-time effects dominate could reflect an averaging artifact: if the pooled coefficient $\hat{\beta}^{\text{pool}}$ is a weighted average of stratum-specific effects, a large $\hat{\beta}^{(n=0)}$ combined with smaller (or negative) $\hat{\beta}^{(n=1)}$ and $\hat{\beta}^{(n \geq 2)}$ would produce a modest pool by construction and the “first break is twice the pool” finding would be mechanical. To document that the ATP first-break excess is not simply arithmetic, Table 7 reports the stratum-weighted decomposition:

$$\hat{\beta}^{\text{pool}} \text{ vs. } \underbrace{w_0 \hat{\beta}^{(n=0)} + w_1 \hat{\beta}^{(n=1)} + w_{\geq 2} \hat{\beta}^{(n \geq 2)}}_{\text{weighted average}},$$

where the weights w_n are the treated-share of each stratum in the pooled sample. Under the law of iterated expectations, the pool coefficient equals the weighted average when treatment effects do not interact with covariates that differ across strata, so any gap between the two is informative about specification.

On the ATP tour, the pool-to-first-LL ratio is 2.17 at 4 weeks, 2.19 at 8 weeks, and 2.07 at 12 weeks: the first-time effect is more than twice the pooled effect, with negative $\hat{\beta}^{(n=1)}$ and $\hat{\beta}^{(n \geq 2)}$ pulling the weighted average below the pool. The gap between the pool and the weighted average on the ATP GS side (75 points at 4–12 weeks) reflects that the small $n = 1$ and $n = 2+$ cells produce imprecise negative estimates, and the pooled regression’s within-stratum information does not translate additively into the weighted-average decomposition. The non-GS ATP ratios (1.6–2.1) confirm the doubling pattern in a larger sample. On the WTA GS side, ratios are below one at short horizons, reflecting that WTA pooled effects are driven by repeat rather than first-time recipients; the WTA non-GS pattern is close to one at short horizons and grows at longer ones. The decomposition therefore supports the first-break interpretation on the ATP tour and identifies the WTA tour as a setting where the first-break premium is smaller and horizon-dependent.

Table 7: Mechanical Decomposition: Pool vs. Stratum-Weighted Average (Ranking Points)

	4w	8w	12w	26w	52w
<i>Panel: GS-ATP</i>					
$\hat{\beta}^{\text{pool}}$	31.9	31.3	38.5	49.2	4.3
Weighted avg. $\sum_n w_n \hat{\beta}^{(n)}$	-43.5	-43.7	-36.1	-17.0	-54.3
Gap (pool – wavg)	75.5	75.0	74.6	66.2	58.6
$\hat{\beta}^{(n=0)}$ (first LL)	69.3	68.8	79.8	70.4	21.4
$\hat{\beta}^{(n=0)} / \hat{\beta}^{\text{pool}}$	2.17	2.19	2.07	1.43	4.95
<i>Panel: NonGS-ATP</i>					
$\hat{\beta}^{\text{pool}}$	12.6	12.1	6.6	24.7	-8.0
Weighted avg. $\sum_n w_n \hat{\beta}^{(n)}$	15.4	16.6	12.7	32.4	11.6
Gap (pool – wavg)	-2.8	-4.5	-6.1	-7.7	-19.6
$\hat{\beta}^{(n=0)}$ (first LL)	20.3	19.7	13.6	44.7	37.8
$\hat{\beta}^{(n=0)} / \hat{\beta}^{\text{pool}}$	1.61	1.63	2.07	1.81	-4.73
<i>Panel: GS-WTA</i>					
$\hat{\beta}^{\text{pool}}$	64.4	78.7	32.3	6.3	6.3
Weighted avg. $\sum_n w_n \hat{\beta}^{(n)}$	76.2	78.7	35.8	7.5	12.6
Gap (pool – wavg)	-11.8	-0.0	-3.6	-1.2	-6.3
$\hat{\beta}^{(n=0)}$ (first LL)	36.1	49.6	21.5	1.3	20.6
$\hat{\beta}^{(n=0)} / \hat{\beta}^{\text{pool}}$	0.56	0.63	0.67	0.20	3.28
<i>Panel: NonGS-WTA</i>					
$\hat{\beta}^{\text{pool}}$	27.4	22.2	23.0	3.1	4.7
Weighted avg. $\sum_n w_n \hat{\beta}^{(n)}$	21.5	15.5	14.7	-6.4	-16.3
Gap (pool – wavg)	6.0	6.7	8.3	9.5	21.0
$\hat{\beta}^{(n=0)}$ (first LL)	17.9	19.1	23.4	20.7	59.4
$\hat{\beta}^{(n=0)} / \hat{\beta}^{\text{pool}}$	0.65	0.86	1.02	6.72	12.70

Notes. $\hat{\beta}^{\text{pool}}$ is the coefficient on D_{ie} from the pooled regression on the full sample. Weighted average is $\sum_n w_n \hat{\beta}^{(n)}$, where w_n is the treated-share of stratum n in the pooled sample and $\hat{\beta}^{(n)}$ is the stratum-specific coefficient from Tables 5–6. Ratio is $\hat{\beta}^{(n=0)} / \hat{\beta}^{\text{pool}}$. Positive ratio above one indicates the first-LL effect exceeds the pooled effect; the paper’s headline claim implies a ratio of roughly two.

7.3 Verified Lottery Subsample and Contamination

The Grand Slam lottery mechanism operates only when withdrawals occur before qualifying completes; post-qualifying withdrawals follow the ranking rule (the highest-ranked qualifying loser receives the slot). Withdrawal timing is not recorded in the data, so the main sample may pool lottery-assigned and ranking-assigned observations. This contamination has a signable direction: under ranking-based assignment, treated players are systematically higher-ranked than controls, biasing ranking-point effect estimates upward rather than toward attenuation. Two features of the analysis partially offset this bias. First, the pre-treatment control vector Z_{ie}^{pre} includes ranking points (linear and quadratic), Elo, and surface Elo, absorbing the observable component of selection. Second, within-event fixed effects restrict comparisons to the top-four pool at a given tournament, where the ranking gap between treated and untreated candidates is mechanically small (at most three ranks).

To assess the residual contamination concern, I construct a *verified lottery* subsample using a data-only classification test. Under the ranking rule, the highest-ranked qualifying loser receives the slot deterministically, so any treated LL whose rank among final-round qualifying losers exceeds one must have been selected by lottery. An event enters the verified subsample if at least one of its LL entries satisfies this criterion. Among first-time LL candidates, this criterion identifies 35 verified ATP events yielding $N = 86$ observations (55 treated, 31 control) and 17 verified WTA events yielding $N = 47$ observations (27 treated, 20 control). Table 8 reports stacked dynamic estimates for this subsample. On the ATP tour, ranking-point gains rise from 33.2 at 4 weeks (not significant) to 51.2 at 12 weeks ($p = 0.061$) and peak at 76.6 at 26 weeks ($p = 0.050$), before fading to 12.3 at 52 weeks; the 26-week effect on the verified subsample is close in magnitude to the headline 26-week estimate (72.6) and remains significant under player-clustered inference. An Elo decline at 52 weeks (-44.2 , $p = 0.098$) aligns with an overseeding channel operating on the longest horizon after ranking gains fade. On the WTA tour, the verified subsample delivers a sharper 8-week ranking-point effect ($+77.5$, $p = 0.005$) than the headline WTA sample ($+55.0$, $p = 0.017$) and $+60.8$ at 4

weeks ($p = 0.090$); WTA Elo gains at 4–8 weeks (+15.4 and +21.5, $p = 0.094$ and $p = 0.051$) also strengthen relative to the headline WTA estimates. The verified subsample is smaller than the headline sample and therefore less powerful at any single horizon, but the point estimates align with the main results in sign and are of comparable magnitude, so main-sample contamination from ranking-assigned entries is not driving the headline conclusion.

To probe the residual identification concern from the other direction, I ran a news-based verification of the 23 first-time GS entries whose rank among qualifying losers was one and whose classification therefore rests on withdrawal-timing information not encoded in the data. Searching tennis-media coverage and Wikipedia tournament pages, I found datable withdrawal information for 10 of the 23 entries (7 at high confidence with named withdrawal dates from tennis-media reporting; 3 at medium confidence where the pattern of a late injury withdrawal was reported without an exact date). Every datable case was a post-qualifying withdrawal that triggered the ranking rule rather than the lottery: Kyrgios at Wimbledon 2023 withdrew on Sunday July 2 (the night before main draw); Alexandrova at Wimbledon 2024 withdrew on Monday July 1 (hours before her first-round match); Raonic at Roland Garros 2021 and again at the US Open 2019; Davidovich Fokina at Wimbledon 2024; Andreescu and Badosa at the US Open 2023; and Tomljanovic at the Australian Open 2023. None of the 23 rank-one entries turned out to be a lottery draw in the news check. The remaining 13 rank-one entries are pre-2015 slams with thin contemporaneous news coverage and stay unresolved; the accumulated evidence from the datable subset makes it likely that most of the unresolved cases are also ranking-rule assignments. The news verification therefore aligns with the F14 rank-based rule: the verified-lottery subsample captures the lottery-assigned events cleanly, and treating the rank-one entries as ambiguous rather than as lottery-verified is the appropriate choice.

Table 8: First-LL Verified Lottery Subsample: Dynamic Effects

	4w	8w	12w	26w	52w
<i>ATP: N = 86 (55 treated, 31 control), 35 events</i>					
Ranking Pts Δ	33.2 (28.0)	39.1 (28.7)	51.2* (26.9)	76.6** (38.4)	12.3 (62.4)
Elo Δ	-6.4 (9.6)	-5.4 (12.2)	-10.5 (11.4)	6.1 (15.4)	-44.2* (26.4)
Main Draws	-0.24 (0.58)	-0.19 (0.53)	0.04 (0.53)	0.57 (0.64)	1.05 (1.59)
Matches 250+	-0.35 (1.10)	-0.17 (1.02)	0.28 (1.01)	1.22 (1.18)	1.22 (3.00)
<i>WTA: N = 47 (27 treated, 20 control), 17 events</i>					
Ranking Pts Δ	60.8* (35.1)	77.5*** (26.2)	26.8 (24.4)	9.6 (63.7)	8.5 (118.5)
Elo Δ	15.4* (9.0)	21.5* (10.7)	16.6 (12.2)	15.8 (15.9)	38.3 (23.2)
Main Draws	0.28 (0.66)	0.47 (0.69)	0.83 (0.73)	0.70 (0.85)	0.26 (2.01)
Matches 250+	-0.13 (1.23)	1.04 (1.02)	1.00 (1.08)	1.59 (1.29)	1.17 (3.42)

Notes. Restricted to events with at least one LL entry whose rank among final-round qualifying losers exceeds one, confirming lottery-based assignment. First-time LL candidates only. Stacked dynamic specification with Grand Slam $\times$ Year fixed effects, horizon fixed effects, and pre-treatment controls. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.4 Pre-Treatment Balance

Under genuine within-pool randomization, balance holds in expectation. Finite-sample imbalance is inevitable: with only four candidates per pool and a binary treatment, any single draw produces deviations that are properly interpreted as sampling variability rather than as evidence against randomization. Controls included in the outcome specification improve precision by reducing residual variance; they do not correct for a violation of random assignment, because under the lottery no such violation is present in expectation.

Table 9 reports formal balance tests on pre-treatment covariates. In the first-time ATP Grand Slam sample ($N = 120$), the joint F -test yields $p = 0.106$; in the first-time WTA sample ($N = 82$), $p = 0.063$. Table 24 reports the within-event balance test, which is the economically relevant specification since randomization operates within each event’s top-four pool: the within-event joint F -test yields $p = 0.046$ for ATP and $p = 0.516$ for WTA. A 5% rejection in one of two tests is within the distribution one would expect under genuine randomization; with seven covariates tested across two tours, the expected number of nominal 5% rejections under the null is 0.7, so observing one marginally significant individual imbalance ($p = 0.034$ on age) is consistent with chance. The ATP within-event imbalance is driven by age, a covariate that cannot plausibly be correlated with lottery draws and that is included in Z_{ie}^{pre} to improve precision. Event study plots (Figures 1 and 2) show parallel trajectories in the weeks preceding each Grand Slam, so systematic pre-treatment divergence is not detectable. Non-GS balance tests are reported in the appendix.

7.5 Control Function Specification Check

The CF specification check (Appendix Table 20) applies the control function to the randomized GS sample using a uniform selection probability $P_i^{LL} = c_e/N_e$. On the WTA tour, the specification passes: $\hat{\rho}_h$ is insignificant at all horizons for ranking points (e.g., $\hat{\rho}_4 = 17.5$, SE 48.5) and treatment effect estimates change modestly. On the ATP tour, the specification fails: $\hat{\rho}_h$ is large and highly significant (e.g., $\hat{\rho}_4 = 90.6$, $p < 0.01$), indicating that the uniform

Table 9: Pre-Treatment Balance: Grand Slam Samples

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.48	0.45	0.088	0.53	0.42	0.007
(Ranking points / 1000) ²	0.25	0.21	0.083	0.31	0.20	0.006
Elo / 100	17.30	17.30	0.977	18.54	17.42	0.022
(Elo / 100) ²	299.27	299.28	0.977	343.91	303.37	0.022
Surface Elo / 100	16.29	16.38	0.686	17.62	17.28	0.407
(Surface Elo / 100) ²	266.91	270.07	0.676	314.28	301.26	0.366
Age	24.93	26.20	0.034	23.15	23.83	0.391
Joint F -test p -value	0.106			0.063		
N	120			82		

Notes. Columns compare means for treated (LL selected) and control (unselected) players at the player-episode level. Covariates include ranking points / 1000 (linear and quadratic), Elo / 100 (linear and quadratic), surface Elo / 100 (linear and quadratic), and age. Prior LL variables excluded (zero by construction for first-time candidates). Joint F -statistic tests the null that all covariates are jointly balanced. First-time ATP: $N = 120$; first-time WTA: $N = 82$. Grand Slam sample, 2006–2024.

probability is a poor approximation for the ATP GS assignment process. This likely reflects that the ATP GS pool size varies substantially across events (from 4 to 10+ candidates depending on the qualifying draw size and withdrawal timing), so the uniform-within-top-4 assumption mischaracterizes the selection. The ATP GS failure does not compromise the lottery-based estimates, which rely only on random assignment within the top-four pool rather than on a functional form assumption. It does indicate that the non-GS CF estimates should be read as corroboration rather than as a free-standing identification strategy, and that the lottery design is the primary basis for causal inference.

7.6 Wildcards

Wildcard entries provide a parallel mechanism for main-draw access. If control players differentially receive wildcards to subsequent events, the counterfactual is contaminated. In the first-LL sample, wildcards are rare among qualifying-level players (ranked 100th–300th), who are generally below the threshold for wildcard consideration at tour-level events. In the 26-week window following the focal event, the average number of wildcard main-draw entries

is 0.08 per treated player and 0.09 per control player on the ATP tour ($p = 0.71$), and 0.12 versus 0.11 on the WTA tour ($p = 0.83$). Differential wildcard receipt is therefore too small to confound the LL treatment effect.

7.7 Statistical Power

Power analysis (Appendix Table 21) shows that the first-LL GS-ATP sample can detect a 5.5 percentage-point shift in match win probability and a 26-week Elo change of approximately 25 points at 80% power. The null Elo results are therefore informative: effects larger than one standard deviation of the within-sample Elo distribution can be ruled out at conventional power levels.

8 Conclusion

A player’s first randomly assigned tournament opportunity produces larger career gains than subsequent ones, and these gains operate through access rather than ability and dissipate within a year. In the first-time ATP Grand Slam sample ($N = 120$), lottery selection raises ranking points by 72.5 at 4 weeks ($p = 0.008$), 82.8 at 12 weeks ($p = 0.002$), and 72.6 at 26 weeks (clustered $p = 0.047$; Fisher $p = 0.164$), fading by 52 weeks as the rolling ranking window absorbs the initial gain. These point estimates are roughly twice the effects estimated on the full sample that pools first-time and repeat LL recipients, and a mechanical decomposition into stratum-weighted components shows that $\hat{\beta}^{(n=0)}/\hat{\beta}^{\text{pool}} \approx 2$ across horizons on the ATP side. On the WTA tour ($N = 82$), effects concentrate at 8 weeks (+55.0, $p = 0.017$), and Elo gains at medium horizons align with a genuine ability channel on the women’s tour. Non-Grand-Slam control-function estimates on larger samples ($N = 2,054$ ATP; $N = 1,423$ WTA) reproduce the ATP access pattern: ranking-point gains of 20.6 at 4 weeks ($p = 0.041$) and 44.6 at 26 weeks ($p = 0.002$). Wald tests reject equality of first-LL and second-LL effects for tournament access ($p = 0.002$ ATP; $p = 0.009$ WTA in the non-GS

sample), and third-plus LL recipients show weak or negative effects.

The dominant mechanism is access accumulation. The ranking system does not adjust for opponent quality: a first-round win at a Grand Slam earns the same points regardless of the opponent’s strength. When a first-time lucky loser earns points at the focal event, the improved ranking grants entry to subsequent tournaments and further chances to accumulate points. The match-level tournament performance model reports $\hat{\delta} = -0.141$ (player-clustered $p = 0.17$) for first-time ATP GS recipients: directionally consistent with overseeding when novice entrants face opponents above their ability level, though the small first-time GS sample does not identify the effect from zero. Ranking-point gains therefore come from more draws entered and more matches played rather than from a per-match performance improvement. The WTA evidence differs modestly, with Elo gains at medium horizons pointing to a first-main-draw learning channel that is absent from the pooled-LL results.

The ranking feedback loop, in which points determine access and access determines point-earning opportunities, amplifies a one-time shock into a multi-month cascade; the cascade is self-limiting because sustained ability dominates a single favorable shock within a year when performance is continuously measured and positions are regularly updated. The comparison with [Oyer \(2006\)](#) and [Oreopoulos et al. \(2012\)](#), where initial conditions cause decade-long career effects in less transparent markets, points to market transparency as the key moderator: the 52-week rolling window provides a natural correction mechanism that other labor markets lack.

Two limitations remain. The Grand Slam lottery sample is small ($N = 120$ ATP, $N = 82$ WTA), limiting power for heterogeneity analysis. Fisher randomization inference, more appropriate than asymptotic inference for this small-sample lottery, supports the main effects at 4–12 weeks but not at 26 weeks, so the middle horizon should be read with appropriate caution. Lottery contamination is an additional caveat: an unknown fraction of observations were assigned by ranking rather than by draw, and ranking-rule assignment selects higher-ranked players. The pre-treatment controls absorb the observable component

of this bias, within-event fixed effects restrict comparisons to narrow ranking gaps, and a verified-lottery subsample built from a data-only criterion (rank-among-losers greater than one) recovers a 26-week ATP ranking-point effect of 76.6 ($p = 0.050$) and an 8-week WTA effect of 77.5 ($p = 0.005$) on 35 ATP and 17 WTA verified events, in line with the headline estimates (Section 7.3). A full resolution of the remaining 23 ambiguous entries would require withdrawal-timing data not available in the public record. Future work could measure longer-run outcomes and test whether the first-break phenomenon generalizes to other tournament-style markets with threshold-based access.

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A Econometric Details

A.1 Match-Level Model

The match-level probability is specified as:

$$p_{itm} = \Lambda(\beta' X_{ijm} + \gamma' Z_{ie}^{pre} + \delta D_{ie} + \rho v_{ie}) \quad (6)$$

where X_{ijm} contains pairwise covariates (ranking points difference / 1000 linear and quadratic, Elo difference / 100 linear and quadratic, surface Elo difference / 100 linear and quadratic, Laplace-smoothed H2H win proportion, H2H encounter count, age difference, and surface indicators) and Z_{ie}^{pre} contains pre-treatment focal player characteristics (ranking points / 1000 linear and quadratic, Elo / 100 linear and quadratic, surface Elo / 100 linear and quadratic, and player age). The trajectory likelihood factorizes into match-level Bernoulli contributions, so estimation reduces to a standard logit on stacked match-level data.

A.2 Control Function

The generalized residual correcting for endogenous LL assignment at non-Grand-Slam events is:

$$\hat{v}_{ie} = D_{ie} \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{P_i^{LL}} - (1 - D_{ie}) \cdot \frac{\phi(\Phi^{-1}(P_i^{LL}))}{1 - P_i^{LL}} \quad (7)$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the standard normal PDF and CDF. Including $\hat{v}_{ie}$ in the outcome equation corrects for selection on unobservables.

A.3 Selection Probability Construction

The selection probability P_i^{LL} is computed from the full combinatorial structure of qualifying match pairings. A weighted logit of match outcomes is estimated on qualifying round matches at non-Grand-Slam LL-granting events, separately for each tour. Given predicted match-

winning probabilities for the $N_e - 1$ other final-round qualifying matches, P_i^{LL} is computed by exact enumeration over all 2^{N_e-1} outcome combinations. When $N_e - 1 > 20$, Monte Carlo simulation with 10,000 draws approximates the probability. For the first-LL-restricted sample, the win model is re-estimated on the restricted event set (ATP: $N = 9,425$ qualifying matches; WTA: $N = 10,009$), and the selection probability is recomputed for the LL-naive pool at each event. The old-vs-new P_i^{LL} correlation is 0.979, confirming stability.

A.4 Inference for the Match-Level Model

Standard errors for the match-level model are cluster-robust at the player level, allowing arbitrary within-player dependence across matches and events. Fisher randomization inference for the ranking-point outcomes (Section 5) is reported alongside the clustered SEs and is preferred for the small Grand Slam samples ($N = 120$ ATP; $N = 82$ WTA).

B Additional Tables and Figures

Table 10: Summary Statistics: ATP Grand Slam (First-LL)

Variable	LL Selected			Not Selected		
	Mean	SD	<i>N</i>	Mean	SD	<i>N</i>
<i>Panel A: Pre-Treatment</i>						
Age	24.93	3.41	61	26.20	3.08	59
Ranking points	482.69	121.82	61	447.24	103.05	59
Elo rating	17.35	0.67	61	17.35	0.67	59
Surface Elo	1629.07	124.59	61	1638.44	128.59	59
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	16.93	62.61	61	-5.44	54.73	59
Ranking points Δ (26w)	83.86	228.02	49	46.67	281.37	46
Ranking points Δ (52w)	45.46	316.89	46	87.46	361.27	37
Elo Δ (4w)	-0.24	25.19	60	0.17	29.63	57
Elo Δ (26w)	22.73	62.54	41	10.37	64.76	43
Elo Δ (52w)	14.66	76.14	40	20.66	87.12	32
Main draws entered (26w)	4.69	3.42	52	4.00	2.71	48
Main draws entered (52w)	10.18	8.07	50	9.54	7.28	41

Notes. First-time LL candidates at ATP Grand Slams, 2006–2024. *N* = 120 (61 treated, 59 control). All players have zero prior LL awards.

Table 11: Summary Statistics: WTA Grand Slam (First-LL)

Variable	LL Selected			Not Selected		
	Mean	SD	<i>N</i>	Mean	SD	<i>N</i>
<i>Panel A: Pre-Treatment</i>						
Age	23.15	3.75	33	23.83	3.16	49
Ranking points	525.70	179.38	33	420.84	147.57	49
Elo rating	19.16	1.13	33	18.59	0.97	49
Surface Elo	1762.44	194.16	33	1728.13	163.39	49
<i>Panel B: Outcomes</i>						
Ranking points Δ (4w)	45.67	68.34	33	21.88	39.93	49
Ranking points Δ (26w)	114.65	222.06	31	126.13	235.35	45
Ranking points Δ (52w)	168.92	432.09	25	149.71	297.64	42
Elo Δ (4w)	7.04	25.40	32	3.61	23.57	48
Elo Δ (26w)	16.51	54.66	24	5.94	54.32	38
Elo Δ (52w)	34.80	82.30	23	4.60	67.33	42
Main draws entered (26w)	4.84	3.03	31	3.89	2.84	47
Main draws entered (52w)	11.90	7.50	29	9.78	6.61	45

Notes. First-time LL candidates at WTA Grand Slams, 2006–2024. $N = 82$ (33 treated, 49 control).

Table 12: Non-Grand-Slam Dynamic Effects: ATP (First-LL, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	20.6** (10.1)	20.5** (10.3)	14.1 (10.1)	44.6*** (14.2)	41.1 (29.9)
Elo Δ	8.4 (6.2)	12.8** (5.4)	11.4** (5.6)	30.8** (12.8)	33.6* (20.3)
Main Draws	-1.4*** (0.2)	-0.9*** (0.2)	-0.3* (0.2)	1.7*** (0.3)	5.2*** (0.6)
Matches 250+	-2.6*** (0.4)	-1.7*** (0.3)	-0.6* (0.3)	3.0*** (0.5)	8.9*** (1.1)
Win %	-0.04 (0.03)	-0.01 (0.02)	-0.00 (0.03)	0.04 (0.03)	0.03 (0.03)

Notes. Control function estimates. First-time ATP non-GS sample ($N = 2,054$, 523 treated), 2007–2024. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 13: Non-Grand-Slam Dynamic Effects: WTA (First-LL, Control Function)

	4w	8w	12w	26w	52w
Ranking Pts Δ	18.1 (11.4)	19.2* (11.1)	23.2* (12.2)	19.2 (17.8)	58.5 (40.1)
Elo Δ	6.2* (3.6)	7.1* (3.8)	10.5** (4.1)	6.2 (5.9)	5.3 (8.2)
Main Draws	-1.1*** (0.2)	-0.7*** (0.1)	-0.4*** (0.1)	0.9*** (0.2)	2.7*** (0.5)
Matches 250+	-2.1*** (0.3)	-1.4*** (0.3)	-0.8*** (0.3)	1.7*** (0.4)	5.6*** (1.1)
Win %	0.01 (0.02)	0.02 (0.02)	0.00 (0.02)	-0.01 (0.02)	-0.01 (0.02)

Notes. Control function estimates. First-time WTA non-GS sample ($N = 1,423$, 378 treated), 2009–2024. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14: Non-Grand-Slam Heterogeneity: ATP (First-LL)

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	19.0*	19.6*	13.4	46.4***	48.5
	(11.0)	(11.3)	(11.3)	(16.2)	(32.1)
$\times$ High	11.9	7.2	5.0	-8.3	-40.4
	(10.9)	(11.2)	(11.7)	(18.7)	(27.3)
Main Draws	-1.7***	-1.1***	-0.4**	2.0***	6.2***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
$\times$ High	2.1***	1.5***	0.9***	-1.4***	-5.7***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
Elo Δ	8.6	12.9**	11.9**	28.8**	33.2
	(6.2)	(5.6)	(6.0)	(13.0)	(20.3)
$\times$ High	-1.7	-1.5	-4.2	12.0	1.7
	(5.9)	(6.3)	(7.2)	(9.8)	(14.1)
<i>Age</i>					
Ranking Pts Δ	26.9**	28.6**	21.3*	51.6***	46.4
	(11.4)	(12.0)	(11.8)	(16.8)	(31.9)
$\times$ High	-18.2	-23.8*	-21.3	-20.7	-16.7
	(11.2)	(13.0)	(15.1)	(23.2)	(39.6)
Main Draws	-1.4***	-0.9***	-0.3*	1.7***	5.1***
	(0.2)	(0.2)	(0.2)	(0.3)	(0.7)
$\times$ High	-0.0	-0.0	0.1	0.1	0.3
	(0.3)	(0.3)	(0.2)	(0.4)	(1.0)
Elo Δ	13.7**	17.0***	16.5***	30.1**	28.5
	(6.6)	(5.9)	(6.1)	(13.2)	(20.3)
$\times$ High	-16.0***	-12.4**	-15.3**	3.0	18.8
	(5.1)	(5.8)	(6.6)	(8.8)	(12.8)

Notes. Interaction specifications by rank and age. First-time ATP non-GS sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: Non-Grand-Slam Heterogeneity: WTA (First-LL)

	4w	8w	12w	26w	52w
<i>Rank</i>					
Ranking Pts Δ	16.5 (14.3)	15.1 (14.3)	22.5 (15.7)	13.0 (22.8)	74.5 (51.4)
$\times$ High	5.8 (14.0)	14.0 (14.3)	2.7 (16.0)	20.4 (23.2)	-57.9 (51.3)
Main Draws	-1.7*** (0.2)	-1.0*** (0.2)	-0.6*** (0.2)	1.5*** (0.3)	4.8*** (0.6)
$\times$ High	2.1*** (0.2)	1.3*** (0.2)	0.7*** (0.2)	-1.7*** (0.3)	-6.4*** (0.6)
Elo Δ	4.2 (3.9)	3.5 (4.2)	6.2 (4.5)	1.3 (6.3)	2.2 (9.3)
$\times$ High	4.9 (4.9)	10.7* (5.4)	12.9** (5.8)	14.1* (8.1)	8.9 (12.4)
<i>Age</i>					
Ranking Pts Δ	14.9 (14.0)	9.1 (14.6)	6.1 (17.0)	-29.1 (22.7)	-34.0 (39.6)
$\times$ High	6.2 (14.2)	18.6 (15.9)	30.6* (18.2)	82.7*** (31.0)	144.4** (58.4)
Main Draws	-1.0*** (0.2)	-0.5*** (0.2)	-0.3* (0.2)	0.9*** (0.3)	1.9*** (0.6)
$\times$ High	-0.2 (0.2)	-0.3 (0.2)	-0.1 (0.2)	0.1 (0.3)	1.3* (0.8)
Elo Δ	7.8* (4.3)	10.5** (4.6)	13.0** (5.0)	7.1 (7.5)	1.1 (11.5)
$\times$ High	-3.0 (4.5)	-6.2 (4.9)	-4.6 (5.4)	-1.9 (7.7)	6.4 (12.7)

Notes. Interaction specifications by rank and age. First-time WTA non-GS sample. Standard errors clustered at the player level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Match-Level Tournament Performance: All Four Samples (First-LL)

	GS-ATP	GS-WTA	NonGS-ATP	NonGS-WTA
$\hat{\delta}$ (LL entry)	-0.141 (0.103)	0.146 (0.138)	0.609*** (0.026)	-0.033 (0.029)
Matches	4,445	3,715	407,959	47,376

Notes. Logit estimates of LL entry on match win probability. Pooled specification with pairwise covariates and focal player controls. First-LL restricted samples. Standard errors clustered at the player level. The non-GS specification includes the control function residual. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 17: Pre-Treatment Balance: Non-Grand-Slam (First-LL)

Variable	ATP			WTA		
	LL	Control	p	LL	Control	p
Ranking points / 1000	0.38	0.25	0.000	0.56	0.39	0.000
(Ranking points / 1000) ²	0.19	0.09	0.000	0.41	0.22	0.000
Elo / 100	17.30	16.97	0.000	18.23	17.58	0.000
(Elo / 100) ²	300.03	289.95	0.000	334.01	311.54	0.000
Surface Elo / 100	16.43	16.33	0.084	18.23	17.87	0.000
(Surface Elo / 100) ²	271.54	267.70	0.053	334.01	321.26	0.000
Age	26.15	25.18	0.000	24.02	23.63	0.108
Joint F -test p -value	0.000			0.000		
N	2054			1423		

Notes. Non-GS sample is not randomized; imbalance is expected and motivates the control function. First-time ATP: $N = 2,054$; First-time WTA: $N = 1,423$.

Table 18: Non-Grand-Slam Immediate Effects (First-LL, Control Function)

Outcome	t -test	CF + Tourney FE	
	LL Mean	$\hat{\beta}$	(SE)
<i>ATP</i>			
Main-draw match win	0.26***	0.39***	(0.02)
Main-draw matches played	1.38***	1.58***	(0.03)
Ranking points at event	0.00	NA	(NA)
N	523 / 1531	2054	
<i>WTA</i>			
Main-draw match win	0.37***	0.38***	(0.02)
Main-draw matches played	1.50***	1.50***	(0.03)
Ranking points at event	0.00	NA	(NA)
N	378 / 1045	1423	
<i>Pooled</i>			
Main-draw match win	0.31***	0.38***	(0.01)
Main-draw matches played	1.43***	1.54***	(0.02)
Ranking points at event	0.00	NA	(NA)
N	901 / 2576	3477	

Notes. Control function estimates at the focal event. First-time LL candidates only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 19: Fisher Randomization Inference (First-LL)

	4w	8w	12w	26w	52w
<i>Panel A: ATP</i>					
Ranking pts Δ	74.953*** (23.601)	70.204** (24.024)	81.324** (25.376)	65.034 (35.995)	12.209 (58.417)
Fisher p	0.003	0.022	0.029	0.164	0.897
Elo Δ	-1.203 (9.004)	1.027 (10.917)	-5.560 (10.451)	4.430 (13.805)	-24.598 (21.792)
Fisher p	0.913	0.945	0.639	0.812	0.293
Main draws entered	-0.046 (0.474)	0.228 (0.457)	0.290 (0.462)	0.626 (0.558)	0.693 (1.485)
Fisher p	0.916	0.653	0.594	0.377	0.630
<i>Panel B: WTA</i>					
Ranking pts Δ	45.813 (30.215)	52.359** (20.379)	16.285 (19.180)	-4.464 (47.169)	4.937 (78.627)
Fisher p	0.120	0.024	0.447	0.934	0.954
Elo Δ	10.575 (7.922)	14.236 (9.125)	8.576 (9.847)	27.849* (13.110)	36.157 (20.343)
Fisher p	0.188	0.194	0.464	0.084	0.133
Main draws entered	0.153 (0.539)	0.398 (0.545)	0.360 (0.600)	0.529 (0.705)	0.742 (1.639)
Fisher p	0.710	0.435	0.563	0.508	0.616
Permutations: 1,000; treatment permuted within slam $\times$ year					
N (ATP)		103 $\times$ 5 horizons			
N (WTA)		77 $\times$ 5 horizons			

Notes. Fisher p -values from 1,000 permutations within Grand Slam $\times$ Year blocks. First-time ATP GS sample. Permutation retains only blocks with at least one treated and one control observation; singleton blocks (single-player events) contribute no permutation variation and are dropped, reducing the effective sample from $N = 120$ to $N = 103$.

Table 20: Control Function Validation on GS Lottery Sample (First-LL)

	4w	8w	12w	26w	52w
<i>ATP</i>					
Ranking Pts Δ	72.5*** (26.7)	72.1*** (26.6)	82.8*** (26.1)	72.6** (36.2)	23.5 (59.1)
+ CF	-106.2* (57.2)	-103.3* (60.4)	-93.0 (60.0)	-52.3 (61.6)	-128.4 (85.6)
$\hat{\rho}_h$					
Elo Δ	-4.4 (9.1)	0.2 (10.9)	-4.8 (10.3)	4.7 (13.5)	-9.7 (21.0)
+ CF	-89.3*** (21.7)	-82.5*** (23.4)	-87.1*** (23.1)	-80.2*** (26.1)	-124.5*** (34.5)
$\hat{\rho}_h$					
Main Draws	-0.2 (0.5)	0.1 (0.5)	0.1 (0.5)	0.7 (0.5)	0.8 (1.5)
+ CF	-2.2* (1.3)	-1.9 (1.3)	-1.8 (1.3)	-0.9 (1.4)	-0.1 (2.2)
$\hat{\rho}_h$					
<i>WTA</i>					
Ranking Pts Δ	40.0 (28.5)	55.0** (22.5)	27.8 (20.1)	7.5 (42.8)	26.5 (86.4)
+ CF	9.9 (79.9)	18.6 (72.8)	26.5 (73.4)	-6.6 (94.2)	-5.4 (133.5)
$\hat{\rho}_h$					
Elo Δ	13.3* (7.6)	15.0* (8.5)	11.8 (9.1)	21.6* (12.7)	36.8* (18.9)
+ CF	24.1 (21.5)	24.8 (23.2)	21.8 (22.8)	26.9 (25.0)	56.7* (32.5)
$\hat{\rho}_h$					
Main Draws	-0.1 (0.5)	0.3 (0.5)	0.3 (0.6)	0.7 (0.7)	1.8 (1.5)
+ CF	1.6 (1.2)	2.0* (1.2)	2.0 (1.2)	2.4* (1.3)	3.0 (2.4)
$\hat{\rho}_h$					

Notes. CF applied to randomized GS sample. Under correct specification, treatment effects are unchanged and $\hat{\rho}_h \approx 0$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 21: Minimum Detectable Effects at 80% Power (First-LL)

	GS-ATP	GS-WTA	NonGS-ATP	NonGS-WTA
<i>Panel A: Match-level performance (δ model)</i>				
$\hat{\delta}$ (log-odds)	—	—	—	—
SE	—	—	—	—
MDE (log-odds, 80% power)	—	—	—	—
MDE (pp win prob)	—	—	—	—
N matches	—	—	—	—
<i>Panel B: Elo change (stacked dynamic model, 26w horizon)</i>				
$\hat{\beta}_{26w}$ (Elo pts)	4.7	21.6	30.7	6.2
SE at 26w	13.5	12.7	12.8	5.9
MDE at 26w (Elo pts)	37.7	35.5	35.8	16.4
<i>Panel C: MDE by horizon (Elo points)</i>				
4w	25.4	21.2	17.4	10.1
8w	30.5	23.8	15.1	10.7
12w	28.8	25.4	15.7	11.5
26w	37.7	35.5	35.8	16.4
52w	58.8	53.0	56.9	22.9

Notes. MDE = $(z_{0.025} + z_{0.80}) \times \sigma / \sqrt{N \times p \times (1 - p)}$, where p is the proportion treated. First-LL GS samples.

Table 22: Economic Magnitudes (First-LL)

	Control Mean	LL Effect	Effect/Mean	Effect/SD
<i>Panel A: ATP Grand Slam (First-LL)</i>				
Ranking pts (baseline)	447	—	—	—
Elo (baseline)	17	—	—	—
Ranking Pts Δ (4w)	-5.4	72.5***	1333.1%	1.325
Ranking Pts Δ (26w)	46.7	72.6**	155.6%	0.258
Ranking Pts Δ (52w)	87.5	23.5	26.8%	0.065
Elo Δ (4w)	0.2	-4.4	-2570.4%	-0.150
Elo Δ (26w)	10.4	4.7	44.9%	0.072
Elo Δ (52w)	20.7	-9.7	-47.1%	-0.112
Main Draws (4w)	0.9	-0.2	-18.5%	-0.195
Main Draws (26w)	4.0	0.7	16.6%	0.245
Main Draws (52w)	9.5	0.8	8.8%	0.115
<i>Panel B: WTA Grand Slam (First-LL)</i>				
Ranking pts (baseline)	421	—	—	—
Elo (baseline)	19	—	—	—
Ranking Pts Δ (4w)	21.9	40.0	182.6%	1.001
Ranking Pts Δ (26w)	126.1	7.5	6.0%	0.032
Ranking Pts Δ (52w)	149.7	26.5	17.7%	0.089
Elo Δ (4w)	3.6	13.3*	367.1%	0.562
Elo Δ (26w)	5.9	21.6*	363.2%	0.397
Elo Δ (52w)	4.6	36.8*	800.9%	0.547
Main Draws (4w)	0.6	-0.1	-22.6%	-0.169
Main Draws (26w)	3.9	0.7	16.9%	0.232
Main Draws (52w)	9.8	1.8	17.9%	0.265
<i>Panel C: ATP Non-Grand Slam (First-LL)</i>				
Ranking pts (baseline)	251	—	—	—
Elo (baseline)	1626	—	—	—
Ranking Pts Δ (4w)	2.8	20.5**	719.4%	0.618
Ranking Pts Δ (26w)	15.0	44.5***	296.5%	0.366
Ranking Pts Δ (52w)	10.0	41.0	410.4%	0.207
Elo Δ (4w)	5.4	8.3	153.8%	0.327
Elo Δ (26w)	11.1	30.7**	276.8%	0.509
Elo Δ (52w)	9.1	33.6*	369.0%	0.454
Main Draws (4w)	0.3	-1.4***	-555.8%	-2.440
Main Draws (26w)	1.6	1.7***	105.3%	0.569
Main Draws (52w)	2.8	5.2***	185.4%	0.926
<i>Panel D: WTA Non-Grand Slam (First-LL)</i>				
Ranking pts (baseline)	390	—	—	—
Elo (baseline)	1842	—	—	—
Ranking Pts Δ (4w)	10.0	18.1	181.2%	0.313
Ranking Pts Δ (26w)	32.1	19.2	59.8%	0.108
Ranking Pts Δ (52w)	32.8	58.5	178.2%	0.199
Elo Δ (4w)	2.0	6.2*	302.3%	0.236
Elo Δ (26w)	12.7	6.2	48.5%	0.101
Elo Δ (52w)	17.6	5.3	30.1%	0.066
Main Draws (4w)	0.5	-1.1***	-203.7%	-1.330
Main Draws (26w)	2.6	0.9***	35.8%	0.271
Main Draws (52w)	4.5	2.7***	59.6%	0.426

Notes. Control group means, treatment effects, and effect-to-mean ratios for key outcomes. First-LL samples.

Table 23: Dose-Response: Grand Slam (First-LL)

	4w	8w	12w	26w	52w
<i>GS-ATP</i>					
Ranking Pts Δ	81.8*** (30.9)	81.7*** (29.7)	91.6*** (28.6)	69.6* (40.5)	5.0 (61.0)
$\times$ Dose	-23.3 (27.3)	-24.2 (27.5)	-20.2 (27.3)	17.6 (28.8)	72.6 (60.3)
Main Draws	0.3 (0.6)	0.5 (0.6)	0.4 (0.6)	0.7 (0.7)	0.3 (1.6)
$\times$ Dose	-1.0** (0.4)	-0.8** (0.4)	-0.5 (0.5)	0.1 (0.6)	2.3 (1.6)
Elo Δ	-1.9 (11.6)	4.3 (12.6)	-2.9 (12.5)	2.3 (15.2)	-10.0 (22.1)
$\times$ Dose	-4.8 (9.5)	-9.2 (10.3)	-3.1 (11.3)	16.5 (20.1)	4.0 (20.5)
<i>GS-WTA</i>					
Ranking Pts Δ	38.6 (33.3)	37.1 (29.2)	10.9 (26.7)	-36.8 (45.9)	-51.0 (84.8)
$\times$ Dose	-38.5 (56.2)	23.4 (44.7)	20.0 (43.0)	126.5*** (33.4)	209.4 (155.7)
Main Draws	0.3 (0.7)	0.6 (0.7)	0.7 (0.7)	0.6 (0.8)	1.2 (1.7)
$\times$ Dose	-1.2 (0.9)	-1.2 (0.9)	-1.4 (1.0)	0.4 (1.3)	2.1 (2.0)
Elo Δ	4.7 (11.2)	4.4 (11.5)	-1.5 (11.3)	9.3 (15.6)	32.6 (23.0)
$\times$ Dose	15.4 (14.9)	23.8* (13.5)	39.1** (16.1)	28.4* (16.5)	3.4 (27.7)

Notes. Treatment interacted with matches won at the LL event. First-LL GS samples. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 24: Within-Event Balance Tests: Grand Slam (First-LL)

Variable	ATP		WTA	
	Coef	p	Coef	p
Ranking points / 1000	0.009	0.599	0.022	0.188
(Ranking points / 1000) ²	0.015	0.371	0.023	0.205
Elo / 100	-0.211	0.144	0.108	0.524
(Elo / 100) ²	-7.192	0.151	3.998	0.531
Surface Elo / 100	-0.154	0.497	-0.146	0.574
(Surface Elo / 100) ²	-5.581	0.448	-4.737	0.607
Age	-1.612	0.040	-0.990	0.285

Notes. Each covariate regressed on treatment indicator with event (Grand Slam $\times$ Year) fixed effects. Joint F -test: ATP $p = 0.046$; WTA $p = 0.516$. First-LL sample.

Table 25: Benjamini-Hochberg q -Values Across Primary Outcomes

Sample		4w	8w	12w	26w	52w
<i>Panel: Ranking points</i>						
GS-ATP	$\hat{\beta}_h$	72.5***	72.1***	82.8***	72.6**	23.5
	p	0.008	0.008	0.002	0.047	0.692
	q (BH)	0.028	0.028	0.009	0.126	0.780
GS-WTA	$\hat{\beta}_h$	40.0	55.0**	27.8	7.5	26.5
	p	0.165	0.017	0.171	0.861	0.760
	q (BH)	0.280	0.054	0.280	0.883	0.822
NonGS-ATP	$\hat{\beta}_h$	20.6**	20.5**	14.1	44.6***	41.1
	p	0.041	0.047	0.166	0.002	0.170
	q (BH)	0.119	0.126	0.280	0.008	0.280
NonGS-WTA	$\hat{\beta}_h$	18.1	19.2*	23.2*	19.2	58.5
	p	0.112	0.085	0.057	0.281	0.145
	q (BH)	0.209	0.174	0.135	0.409	0.264
<i>Panel: Elo</i>						
GS-ATP	$\hat{\beta}_h$	-4.4	0.2	-4.8	4.7	-9.7
	p	0.626	0.986	0.639	0.731	0.644
	q (BH)	0.745	0.986	0.745	0.812	0.745
GS-WTA	$\hat{\beta}_h$	13.3*	15.0*	11.8	21.6*	36.8*
	p	0.084	0.082	0.195	0.093	0.055
	q (BH)	0.174	0.174	0.306	0.182	0.134
NonGS-ATP	$\hat{\beta}_h$	8.4	12.8**	11.4**	30.8**	33.6*
	p	0.178	0.018	0.042	0.016	0.099
	q (BH)	0.284	0.054	0.119	0.054	0.189
NonGS-WTA	$\hat{\beta}_h$	6.2*	7.1*	10.5**	6.2	5.3
	p	0.087	0.063	0.011	0.293	0.517
	q (BH)	0.174	0.144	0.038	0.412	0.667
<i>Panel: Main draws</i>						
GS-ATP	$\hat{\beta}_h$	-0.2	0.1	0.1	0.7	0.8
	p	0.759	0.896	0.777	0.227	0.567
	q (BH)	0.822	0.907	0.829	0.343	0.721
GS-WTA	$\hat{\beta}_h$	-0.1	0.3	0.3	0.7	1.8
	p	0.813	0.618	0.652	0.331	0.247
	q (BH)	0.845	0.745	0.745	0.457	0.366
NonGS-ATP	$\hat{\beta}_h$	-1.4***	-0.9***	-0.3*	1.7***	5.2***
	p	0.000	0.000	0.076	0.000	0.000
	q (BH)	0.000	0.000	0.170	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-1.1***	-0.7***	-0.4***	0.9***	2.7***
	p	0.000	0.000	0.008	0.000	0.000
	q (BH)	0.000	0.000	0.028	0.000	0.000
<i>Panel: Matches 250+</i>						
GS-ATP	$\hat{\beta}_h$	0.3	0.7	0.9	1.5	1.3
	p	0.799	0.462	0.340	0.169	0.650
	q (BH)	0.841	0.606	0.460	0.280	0.745
GS-WTA	$\hat{\beta}_h$	-0.5	0.8	0.6	1.9*	3.2
	p	0.588	0.288	0.428	0.055	0.205
	q (BH)	0.735	0.412	0.571	0.134	0.316
NonGS-ATP	$\hat{\beta}_h$	-2.6***	-1.7***	-0.6*	3.0***	8.9***
	p	0.000	0.000	0.055	0.000	0.000
	q (BH)	0.000	0.000	0.134	0.000	0.000
NonGS-WTA	$\hat{\beta}_h$	-2.1***	-1.4***	-0.8***	1.7***	5.6***
	p	0.000	0.000	0.006	0.000	0.000
	q (BH)	0.000	0.000	0.024	0.000	0.000

Notes. BH correction applied jointly across the family of 80 tests (4 primary outcomes $\times$ 4 samples $\times$ 5 horizons). Primary outcomes are ranking points, Elo, main draws entered, and matches at 250+ level events. $q < 0.05$ indicates significance after the joint correction. First-LL samples.

Table 26: Prize Money Effect of First-Time LL Entry (USD)

Sample	Immediate (USD)	Dynamic 26w (USD)	Total (USD)	Treated N
GS-ATP	\$93,770***	\$7,260	\$101,030	61
GS-WTA	\$86,667***	\$7,260	\$93,927	33
NonGS-ATP	\$9,000	\$18,810	\$27,810	523
NonGS-WTA	\$6,000	\$10,230	\$16,230	378

Notes. Immediate effect: difference in focal-event prize money between treated (LL recipients, earning based on main draw rounds won) and control (qualifying-round stipend). Dynamic 26w: estimated access gains (additional main draws entered by 26 weeks) evaluated at an average forward-looking first-round prize of \$11,000. Prize figures are approximate 2020–2024 averages; exact values vary by tournament tier and year. First-LL samples. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$ on immediate effect t -test.

Table 27: Dose-Response: Non-Grand-Slam (First-LL)

	4w	8w	12w	26w	52w
<i>NGS-ATP</i>					
Ranking Pts Δ	9.3 (10.1)	1.1 (10.7)	-8.4 (10.7)	17.6 (15.1)	12.1 (32.2)
$\times$ Dose	22.6** (9.5)	37.0*** (8.5)	42.9*** (9.7)	52.4*** (18.0)	56.1* (29.3)
Main Draws	-1.2*** (0.2)	-0.8*** (0.2)	-0.3* (0.2)	1.4*** (0.3)	4.2*** (0.7)
$\times$ Dose	-0.3** (0.2)	-0.2 (0.1)	0.1 (0.1)	0.6** (0.3)	1.9*** (0.6)
Elo Δ	5.8 (6.4)	10.0* (5.7)	8.1 (5.9)	32.1** (13.5)	35.1* (21.3)
$\times$ Dose	4.6** (2.3)	5.0* (2.7)	6.2* (3.3)	-2.4 (4.7)	-2.4 (5.5)
<i>NGS-WTA</i>					
Ranking Pts Δ	3.9 (12.5)	0.9 (12.4)	5.4 (13.9)	7.9 (20.3)	45.8 (44.5)
$\times$ Dose	30.0*** (7.9)	37.2*** (8.5)	37.4*** (9.6)	25.0 (16.3)	33.1 (36.5)
Main Draws	-1.0*** (0.2)	-0.6*** (0.1)	-0.3** (0.1)	0.7*** (0.2)	2.2*** (0.6)
$\times$ Dose	-0.1 (0.1)	-0.1 (0.1)	-0.0 (0.1)	0.4** (0.2)	1.3** (0.6)
Elo Δ	3.7 (3.8)	5.3 (4.0)	9.0** (4.4)	6.7 (6.5)	10.8 (9.2)
$\times$ Dose	4.7* (2.4)	3.4 (2.6)	3.0 (3.4)	-1.0 (5.4)	-12.6 (9.5)

Notes. Treatment interacted with matches won at the LL event. First-LL non-GS samples with CF. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.